

From Repression to Saturation: A Dynamical Systems Theory of Motivational Structures Under Variable Constraints

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Abstract

This paper develops a transdisciplinary mathematical framework for the dynamics of desire, motivation, and drive under variable environmental constraints. Rather than treating desire as a metaphor or a clinical category, we formalize it as a gradient-driven process within a motivational state space governed by constraint functions, stimulus fields, and coherence measures. The framework introduces several inter-related formal models: *directed motivational density* as a measure of the system's capacity to concentrate arousal around a stable object; a *sublimation operator* modeled as entropy reduction over motivational configurations; a *stimulus saturation law* grounded in receptor desensitization kinetics; and a *co-regulation model* of interpersonal intimacy based on coupled oscillator synchronization. A central result is the identification of four qualitative regimes—repression, coherent desire, saturation, and anhedonic collapse—organized in a phase diagram parametrized by stimulus accessibility and normative constraint density. The general principle unifying these models is that desire requires a potential difference: the existence of a gradient in the motivational field, without which directional motivation dissipates. The framework is shown to admit three complementary formal representations—as Ohmic flow, as free-energy minimization, and as gradient descent on a Lyapunov landscape—establishing the structural universality of the underlying principle. A dedicated section applies the formalism to the digital environment, modeled as a regime of instantaneous discharge, infinite stimulus multiplicity, and absent feedback coupling. A meta-theoretical corollary demonstrates that repression-centered theories of motivation, including classical psychoanalysis, are structurally predictable products of high-constraint cultural regimes, thereby situating Freudian theory as a special case within the present framework. The paper is written with the explicit aim of accessibility to researchers from psychology, sociology, and biology, with thorough explanatory passages accompanying every formal element.

Keywords: dynamical systems, desire, motivation, psychoanalysis, sublimation, stimulus saturation, coupled oscillators, phase transitions, digital environment, entropy, self-organization

Contents

1	Introduction	4
2	Prior Formalization of Motivational Phenomena: A Critical Review	5
2.1	Lewinian Field Theory and Topological Psychology	5
2.2	Freudian Metapsychology and Its Formal Interpreters	6
2.3	Nonlinear Dynamical Systems in Psychology	6
2.4	Computational Neuroscience of Reward and Motivation	7
2.5	Interpersonal Neurobiology and Synchronization	7
2.6	Drive-Reduction Theory and Its Descendants	7
2.7	Opponent-Process Theory and Allostatic Models	8
2.8	Embodied and Enactive Cognition	9
2.9	Summary of the Gap	9
3	Axiomatic Foundations	9
4	Directed Motivational Density	12
4.1	Toy Model: Coherence on a One-Dimensional State Space	13
5	The Potential-Difference Principle	14
5.1	The General Principle	14
5.2	Ohmic Representation	15
5.3	Thermodynamic Representation	15
5.4	Lyapunov Landscape Representation	16
5.5	Toy Model: Double-Well Potential on $\mathbb{R}^1$	17
6	Sublimation as Entropy Reduction	18
6.1	Toy Model: Sublimation on a Discrete State Space	20
7	The Stimulus Saturation Law	21
7.1	Toy Model: Hill Response Curves and Desensitization	23
8	Phase Diagram of Motivational Regimes	24
8.1	Toy Model: Regime Transitions in a Reduced One-Dimensional System . .	27
9	Co-Regulation and Coupled Oscillator Synchronization	28
9.1	The Kuramoto Model of Motivational Coupling	28
9.2	Toy Model: Two Oscillators With and Without Coupling	30
10	The Digital Environment as a Constraint Regime	31
10.1	Formal Characterization	31
10.2	Consequences	32
10.3	The Digital Paradox	33
11	Meta-Theoretical Corollary: Observer–Environment Coupling	33

12 Implications for Artificial Intelligence	34
12.1 Artificial Intelligence as Gradient Process	34
12.2 The Flat-Landscape Problem in Artificial and Biological Systems	34
12.3 Information Entropy and Computational Sublimation	35
12.4 Finite Computational Capacity and Motivational Coherence	35
12.5 Synthesis: Motivation as Landscape Architecture	36
13 Universality Across Scales: From Bacteria to Astrophysics	37
13.1 Animal Motivation: Instincts as Fixed Landscape Topography	37
13.2 The Bacterial Limit: Chemotaxis as Minimal Desire	38
13.3 Extraterrestrial Intelligence: The Universality of the Gradient Principle . .	38
13.4 Astrophysical Parallels: Gradient Dynamics at Cosmic Scales	39
14 Discussion	40
14.1 Summary of Results	40
14.2 Relation to Existing Work	40
14.3 Empirical Predictions	40
14.4 Counterintuitive Predictions, Falsifiability, and Scope Boundaries	41
14.4.1 Counterintuitive Predictions	41
14.4.2 Falsifiability Conditions	42
14.4.3 Beyond Gradient Descent: Compulsive Behavior, Reactance, and Oscillatory Dynamics	43
14.5 Limitations and Future Directions	44
14.5.1 On the Conservation Axiom: Toward a Time-Varying $E_{\max}(t)$. . .	44
14.5.2 Additional Limitations	45
14.6 Relation to a Broader Program: Unified Structural Theory of Complex Systems	45
14.7 Concluding Remarks	46
A Practical Intervention Protocols Derived from the Framework	50
A.1 Managed Deficit Protocols: Restoring Gradient Structure in the Saturation Regime	50
A.2 Stimulus Field Differentiation: Combating Motivational Diffusion	51
A.3 Interpersonal Synchronization Exercises	51
A.4 Sublimation as Cognitive Architecture: Channeling Undischarged Energy .	51
B Self-Assessment Tool: Phase Regime Identification	52
B.1 Stimulus Accessibility Assessment	52
B.2 Normative Constraint Assessment	53
B.3 Coherence Indicator	53
B.4 Interpretation Guide	54

1 Introduction

The scientific study of desire has long been fragmented across disciplinary boundaries. Psychology treats desire as drive or affect; neuroscience locates it in dopaminergic circuits; sociology frames it through norms and taboos; philosophy interrogates its intentional structure. What has been absent is a formal, mathematically grounded theory that captures the *architecture* of desire itself—the structural conditions under which motivational states arise, concentrate, transform, or dissolve—independent of any single disciplinary ontology.

Classical psychoanalysis, inaugurated by Freud, offered the most ambitious prior attempt at a structural account. Freud described an internal economy of psychic energy, positing that desire (“libido”) arises from biological drive, meets resistance in the form of social prohibition, and is either discharged, repressed, or sublimated [Freud, 1905, 1930]. This tripartite model was productive but limited: it was calibrated to a specific cultural regime characterized by strong normative prohibition and restricted stimulus access. The model explained repression well because repression was the dominant regulatory mode of its environment.

The present century has witnessed a fundamental change in the constraint landscape governing desire. The shift is not from prohibition to freedom but from *scarcity to saturation*. In the digital environment, erotic and motivational stimuli are available instantaneously, in unlimited multiplicity, and without the reciprocal feedback loops that characterize embodied interaction. The result is not liberation from the Freudian conflict but a *topological transformation* of it: the pathology of the 21st century is not neurosis born of repression but a diffuse incapacity born of overstimulation.

This paper proposes a mathematical framework adequate to both regimes and, more importantly, to the general structural principles underlying any regime of desire. We formalize desire not as a substance or energy in the metaphorical sense but as a *gradient-driven dynamical process* within a motivational state space. The core constructs are:

1. A **motivational state space** $\mathcal{M}$ in which the system’s current configuration of drives and orientations is represented as a point or distribution.
2. A **stimulus field** $\mathcal{S}$ representing the external landscape of available objects, signals, and incentives.
3. A **constraint function** $\mathcal{C}$ encoding the normative, social, and biological restrictions imposed on the system.
4. A **directed motivational density** ρ measuring the system’s capacity to concentrate arousal around stable objects.
5. A **motivational potential** Φ whose gradient drives the system’s trajectory through $\mathcal{M}$.

From these primitives, we derive a coherent family of models—sharing common vocabulary and axioms—that address sublimation, saturation, co-regulation, and regime transitions. The unifying thesis is that *desire requires a potential difference*: without a gradient in the motivational landscape, the system loses directionality and collapses into either stasis or diffusion.

The paper is organized as follows. Section 2 provides a critical review of prior attempts to formalize motivational and affective phenomena mathematically. Section 3 establishes

the axiomatic foundations. Section 4 introduces directed motivational density and the coherence measure. Section 5 develops the potential-difference principle in three complementary formalisms. Section 6 formalizes sublimation as an entropy-reducing transformation. Section 7 derives the stimulus saturation law with pharmacological parallels. Section 8 constructs the phase diagram with four named regimes. Section 9 models co-regulation as coupled oscillator synchronization. Section 10 applies the framework to the digital environment. Section 11 presents the meta-theoretical corollary on observer-environment coupling. Section 12 explores implications for artificial intelligence. Section 13 examines the framework’s universality across scales, from bacterial chemotaxis to astrophysical dynamics. Section 14 discusses implications and future directions.

2 Prior Formalization of Motivational Phenomena: A Critical Review

The attempt to bring mathematical rigor to the study of motivation, desire, and affect has a long and instructive history. This section surveys the principal programs of formalization, identifies their structural contributions and limitations, and establishes the lacunae that the present framework aims to fill.

2.1 Lewinian Field Theory and Topological Psychology

The earliest sustained attempt to mathematize motivational dynamics was Kurt Lewin’s field theory, developed across a series of publications from the 1930s to the late 1940s [Lewin, 1936, 1938, 1951]. Lewin proposed that behavior is a function of the person and the environment, $B = f(P, E)$, and sought to represent the “life space”—the totality of an individual’s psychological field—using concepts from topology. In his framework, motivational states correspond to regions in a topological space; goals and aversions carry positive or negative *valences*; movement through the space is governed by *vectors* representing psychological forces; and *barriers* separate accessible from inaccessible regions.

Lewin’s conceptual contributions were profound. The idea that motivation is a field phenomenon—that desire arises not from an isolated internal state but from the structural relationship between an agent and its environment—anticipates the present framework’s treatment of desire as a gradient in a potential landscape. His concept of “tension systems” created by unfulfilled needs and discharged by goal attainment is a direct precursor to our axiom of gradient dynamics (Axiom 3.1).

However, Lewin’s mathematical implementation was widely recognized as inadequate to his conceptual ambitions. As numerous commentators have observed, his use of topology remained more metaphorical than formal: the topological spaces he drew lacked the metric structure needed to define distances, gradients, and rates of change. He was, by his own admission, unaware of how geometries beyond simple topology could be given psychological content [Rummel, 1975]. The present framework addresses this gap directly by equipping the motivational state space with a smooth manifold structure, a potential function with well-defined gradients, and explicit dynamical equations.

2.2 Freudian Metapsychology and Its Formal Interpreters

Freud’s metapsychological writings, particularly the “Project for a Scientific Psychology” (1895) and “Beyond the Pleasure Principle” (1920), employed hydraulic and thermodynamic metaphors to describe the flow and accumulation of psychic energy [Freud, 1895, 1920]. These metaphors—tension, discharge, cathexis, binding—constituted an implicit dynamical systems theory, but Freud never formalized them mathematically.

Several later investigators attempted this formalization. Rapaport’s systematization of psychoanalytic metapsychology in the 1960s organized Freud’s concepts into a quasi-axiomatic structure but did not introduce mathematical equations [Rapaport, 1960]. More recently, Friston’s free-energy principle has been proposed as a rigorous computational framework that subsumes aspects of Freudian theory, particularly the pleasure principle (interpreted as surprise minimization) and the notion of bound energy (interpreted as precision-weighted prediction error) [Friston, 2010, Carhart-Harris and Friston, 2010]. Hopkins and others have argued that the free-energy principle provides the missing formalization of Freud’s “Project” [Hopkins, 2012].

Our framework relates to but differs from these efforts in a specific way. While the free-energy principle provides a powerful and general account of perception, action, and learning under uncertainty, its treatment of motivation and desire is indirect: desire is implicit in the prior preferences that generate prediction errors. The present framework makes desire the primary object of formalization, introducing constructs—directed motivational density, the sublimation coefficient, the saturation threshold—that have no direct counterparts in the free-energy literature. In this sense, the two programs are complementary rather than competing.

2.3 Nonlinear Dynamical Systems in Psychology

The application of nonlinear dynamical systems theory to psychological phenomena has been a growing program since the 1990s. The landmark work of Thelen and Smith on developmental dynamics demonstrated that behavioral transitions in infants could be modeled as phase transitions in coupled dynamical systems, with motor milestones emerging from self-organization rather than from pre-specified genetic programs [Thelen and Smith, 1994]. Vallacher and Nowak applied dynamical systems to social psychology, modeling attitude change, interpersonal synchronization, and social coordination as attractor dynamics [Vallacher and Nowak, 1994, 2002]. Guastello and colleagues compiled the field’s diverse applications in their comprehensive volume on chaos and complexity in psychology [Guastello et al., 2009].

In clinical domains, Gottman and colleagues pioneered the use of coupled differential equations to model marital interaction, treating partners as mutually influencing dynamical systems with repair and damage thresholds [Gottman et al., 2002]. Schiepek developed dynamical models of psychotherapy that track problem intensity, motivation, and emotion as interacting state variables with characteristic phase transitions at moments of therapeutic breakthrough [Schiepek et al., 2016]. Tschacher and colleagues applied the Fokker–Planck equation to model the probability distribution of patient states across therapy [Tschacher et al., 2015].

These contributions established the legitimacy and productivity of dynamical systems approaches in psychology. However, they share a common limitation from the perspective of the present work: they are domain-specific models—models of marital conflict, infant locomotion, therapeutic process—rather than general theories of the architecture of desire.

The present framework draws on these technical precedents (gradient dynamics, attractor landscapes, phase transitions) while aiming at a higher level of generality: not a model of any particular manifestation of desire, but a formal theory of the structural conditions under which desire of any kind arises, concentrates, transforms, or dissipates.

2.4 Computational Neuroscience of Reward and Motivation

The computational neuroscience of reward has produced a rich formal literature grounded in temporal-difference learning, dopaminergic prediction error signals, and optimal control theory [Schultz et al., 1997, Sutton and Barto, 1998, Dayan and Niv, 2008]. In these models, motivation is operationalized as the expected value of future reward, and the agent’s policy is optimized to maximize cumulative discounted return. Berridge and Robinson’s incentive salience theory introduced a critical distinction between “wanting” (incentive motivation, mediated by dopaminergic systems) and “liking” (hedonic pleasure, mediated by opioidergic systems), demonstrating that the two can be dissociated [Berridge and Robinson, 2003, Berridge and Kringelbach, 2015].

The pharmacological dimensions of our saturation model (Section 7) are directly informed by this literature, particularly the evidence on dopaminergic habituation and receptor downregulation under chronic stimulation [Volkow et al., 2011]. However, computational reward models are typically concerned with optimal decision-making under uncertainty, not with the structural dynamics of desire itself. They do not address the question that is central to the present framework: what happens to the *architecture* of motivation when the constraint landscape changes qualitatively—when the environment shifts from scarcity to saturation, from prohibition to permissiveness?

2.5 Interpersonal Neurobiology and Synchronization

The neurobiology of interpersonal synchronization has become a rapidly growing field. Hasson and colleagues demonstrated brain-to-brain coupling during communication using fMRI hyperscanning [Hasson et al., 2012]. Dumas and colleagues showed inter-brain synchronization during social interaction using dual-EEG recordings [Dumas et al., 2010]. Feldman’s extensive work on the neurobiology of attachment has documented the role of oxytocin, cortisol, and autonomic nervous system co-regulation in the formation of parent–infant and romantic bonds [Feldman, 2012, 2017].

The Kuramoto model, which we employ in Section 9, is well established in physics and neuroscience as a canonical model of synchronization in coupled oscillator networks [Kuramoto, 1984, Strogatz, 2000]. Its application to interpersonal affective synchronization has been explored by Raz and colleagues, who used Kuramoto-like models to describe the entrainment of parent–infant physiological rhythms [Raz et al., 2012]. Our contribution is to embed this synchronization model within the broader motivational framework, deriving the distinction between deep satisfaction and surface arousal as a formal consequence of coupling strength.

2.6 Drive-Reduction Theory and Its Descendants

Clark Hull’s drive-reduction theory, formalized in his *Principles of Behavior* [Hull, 1943], represents one of the earliest systematic attempts to cast motivation in mathematical terms. Hull proposed that behavior is a multiplicative function of drive (a general energizing state arising from biological need) and habit strength (learned stimulus–response

associations), yielding the equation $sE_R = sH_R \times D \times K$, where D is drive, sH_R is habit strength, and K is incentive motivation. The theory predicted that motivation increases monotonically with deprivation and decreases with satiation—a drive-reduction mechanism.

Hull’s framework shares with the present theory the commitment to treating motivation as a quantitative, formally specifiable process and the insight that biological need generates a scalar intensity (analogous to our potential Φ). However, Hull’s model is *scalar*: drive is a single number, not a distribution over a state space, and the theory cannot represent the spatial structure of motivation—the difference between focused and diffuse desire that is central to our coherence measure Λ . More critically, Hull’s drive-reduction principle predicts that organisms always seek to minimize arousal, which fails to account for sensation-seeking, curiosity, and the intensification of desire under moderate constraint. The present framework supersedes drive-reduction by treating arousal and satisfaction as emergent properties of the potential landscape’s geometry rather than as scalar quantities to be minimized.

2.7 Opponent-Process Theory and Allostatic Models

Solomon and Corbit’s opponent-process theory [Solomon and Corbit, 1974] proposed that every affective state automatically triggers an opposing process: pleasure triggers a delayed pain response, and vice versa. With repeated exposure, the opponent process (the “b-process”) strengthens while the primary process (the “a-process”) remains constant, producing tolerance to the primary affect and a growing aftereffect of the opposite sign. This mechanism accounts for phenomena ranging from drug tolerance to the intensification of grief after loss.

The opponent-process model relates directly to the desensitization dynamics of Section 7. The strengthening of the b-process under repeated exposure is formally analogous to the rightward shift of the Hill-function parameters ($K_d \rightarrow K_d(1 + \gamma)$) in Proposition 7.2: both describe a system that progressively requires stronger stimuli to achieve the same net response. However, the opponent-process model is confined to the temporal dynamics of a single stimulus–response pair, whereas the present framework embeds desensitization within a spatial, multi-object motivational landscape. The key extension is from “how does the response to *this* stimulus change over time?” to “how does the entire architecture of desire reorganize when the stimulus field changes?”

Sterling’s allostatic model of motivation [Sterling, 2012] extends the opponent-process idea by proposing that the organism does not merely react to perturbations but actively predicts future demands and adjusts its set points accordingly. In allostasis, the “baseline” itself is a dynamic variable. This is closely related to our treatment of the intrinsic potential U_{int} as a structure that can be reshaped by chronic environmental conditions—the potential landscape is not fixed but adapts to the regime it inhabits. The allostatic perspective also provides neurobiological support for the transition from Regime III to Regime IV: chronic allostatic overload, in which the system’s predictive adjustments become maladaptive, corresponds to the progressive degradation of $\mathcal{R}_{\text{max}}$ that drives the system into anhedonic collapse. Koob and Le Moal’s allostatic model of addiction [Koob and Le Moal, 2001] makes this mechanism explicit for substance use disorders, and Section 14.4 discusses how the present framework accommodates compulsive behavior through landscape distortion.

2.8 Embodied and Enactive Cognition

The present framework’s emphasis on co-regulation as mutual synchronization of embodied systems (Section 9) intersects with the embodied and enactive traditions in cognitive science. Varela, Thompson, and Rosch’s foundational work on enaction [Varela et al., 1991] argued that cognition is not disembodied computation but emerges from the structural coupling between an organism and its environment. Thompson’s later development of this program [Thompson, 2007] emphasized that affect and motivation are constitutively embodied: they arise from the self-organizing dynamics of the living body in interaction with its surroundings.

The enactive framework provides philosophical grounding for two features of the present theory. First, our treatment of co-regulation as Kuramoto coupling between embodied oscillators (not as information exchange between disembodied minds) is consistent with the enactive emphasis on intercorporeality—the idea that interpersonal understanding is achieved through bodily coordination rather than mental simulation [Gallagher, 2005]. Second, the axiom that motivational capacity $E_{\max}$ is bounded by the biological substrate (Axiom 3.4) reflects the enactive commitment that cognition is constrained by the organism’s material constitution. The present framework adds to the enactive program a precise mathematical formalism—gradient dynamics, Lyapunov stability, entropy functionals—that the enactive tradition has generally lacked, while the enactive tradition provides the framework with a philosophically articulated account of why embodiment matters for motivation.

2.9 Summary of the Gap

The preceding review reveals a landscape of partial formalizations, each capturing an important aspect of motivational dynamics but none providing a unified, axiomatically grounded, regime-sensitive theory of desire. Lewin provided the conceptual architecture but lacked the mathematical tools. Freud provided the phenomenological richness but not the formalism. Hull quantified drive but could not represent motivational geometry. Solomon and Corbit captured temporal opponent dynamics but not spatial reorganization. Sterling formalized allostatic adaptation but not the full regime structure. The nonlinear dynamics program provided the technical precedents but not the motivational specificity. Computational neuroscience provided the reward-theoretic framework but not the concern with structural transitions. The enactive tradition provided the philosophical grounding for embodied motivation but not the mathematical apparatus. Interpersonal neurobiology provided the synchronization evidence but not the embedding within a general theory of desire.

The present framework aims to occupy this gap: a mathematically rigorous, axiomatically grounded, modular theory of desire dynamics that is general enough to encompass multiple regimes (repression, coherence, saturation, collapse) and specific enough to generate testable predictions about each.

3 Axiomatic Foundations

We begin by establishing the primitive concepts and axioms from which all subsequent models are constructed. The goal is to provide a minimal but sufficient formal vocabulary that can serve as common ground across the distinct models developed in later sections.

The motivational state space is the arena in which all dynamics unfold. It represents the totality of possible configurations of an agent’s drives, orientations, and arousal patterns at a given moment. Each point in this space corresponds to a particular distribution of motivational intensity across possible objects of desire.

Definition 3.1 (Motivational State Space). The *motivational state space* is a smooth, finite-dimensional manifold $\mathcal{M} \subseteq \mathbb{R}^n$, where n represents the number of independent motivational dimensions available to the agent. A *motivational state* at time t is a point $\mathbf{m}(t) \in \mathcal{M}$.

The dimensionality n captures the degrees of freedom of the motivational system. In a simple model, these might correspond to intensity of attachment toward distinct categories of objects; in a richer model, they could include temporal discounting preferences, arousal levels, and orientation parameters. The manifold structure ensures that nearby states are meaningfully similar and that smooth transitions between states are possible.

Definition 3.2 (Stimulus Field). The *stimulus field* is a time-dependent, non-negative function $\mathcal{S} : \mathcal{M} \times [0, \infty) \rightarrow \mathbb{R}_{\geq 0}$ that assigns to each point in motivational space and each moment in time a scalar value representing the density and salience of externally available stimuli relevant to that motivational configuration.

The stimulus field is the environment as seen from the perspective of the motivational system. A high value of $\mathcal{S}(\mathbf{m}, t)$ at a particular point $\mathbf{m}$ means that stimuli resonant with the motivational configuration $\mathbf{m}$ are abundantly and saliently available at time t . In a pre-digital society, the stimulus field for erotic motivations might be sparse and highly localized; in a digital environment, it becomes dense, homogeneous, and persistent.

Definition 3.3 (Constraint Function). The *constraint function* is a time-dependent mapping $\mathcal{C} : \mathcal{M} \times [0, \infty) \rightarrow \mathbb{R}_{\geq 0}$ that assigns to each motivational state and time a scalar representing the cumulative normative, social, institutional, and biological cost associated with occupying or transitioning through that state.

The constraint function encodes what Freud conceptualized as the superego and the reality principle, but in generalized form: it includes not only moral prohibition but also social stigma, institutional regulation, biological refractory limits, and opportunity costs. A high value of $\mathcal{C}(\mathbf{m}, t)$ makes the state $\mathbf{m}$ costly to occupy; the system experiences resistance when moving toward or through such states. We define the *normative density* at a point as the magnitude of the constraint gradient:

Definition 3.4 (Normative Density). The *normative density* at a point $\mathbf{m} \in \mathcal{M}$ at time t is

$$\nu(\mathbf{m}, t) = \|\nabla_{\mathbf{m}} \mathcal{C}(\mathbf{m}, t)\|, \quad (1)$$

where $\nabla_{\mathbf{m}}$ denotes the gradient with respect to the motivational coordinates.

A region of high normative density is one in which the cost landscape is steep—small changes in motivational state produce large changes in normative cost. This corresponds to cultural environments with rigid taboos surrounding particular desires. A region of low normative density is permissive: the cost landscape is flat and offers little resistance.

We now state the core axioms of the framework.

Axiom 3.1 (Gradient Dynamics). The motivational state evolves according to a gradient flow driven by a motivational potential $\Phi : \mathcal{M} \times [0, \infty) \rightarrow \mathbb{R}$:

$$\frac{d\mathbf{m}}{dt} = -\nabla_{\mathbf{m}}\Phi(\mathbf{m}, t) + \boldsymbol{\xi}(t), \quad (2)$$

where $\boldsymbol{\xi}(t)$ is a stochastic noise term representing endogenous fluctuations in drive.

This axiom asserts that the motivational system moves in the direction that decreases its potential most rapidly, subject to random perturbations. This is analogous to a particle descending a potential energy landscape in physics: the system flows “downhill” toward states of lower motivational tension. The noise term $\boldsymbol{\xi}(t)$ accounts for the spontaneous fluctuations in drive that are ubiquitous in biological systems—momentary shifts in attention, hormonal variations, stochastic neural firing.

Axiom 3.2 (Potential Composition). The motivational potential decomposes as

$$\Phi(\mathbf{m}, t) = -\mathcal{S}(\mathbf{m}, t) + \mathcal{C}(\mathbf{m}, t) + U_{\text{int}}(\mathbf{m}), \quad (3)$$

where $U_{\text{int}}(\mathbf{m})$ is an intrinsic potential encoding the system’s internal drive structure.

The potential has three components. The stimulus field $\mathcal{S}$ appears with a negative sign because the presence of resonant stimuli *attracts* the system (lowers the potential); the constraint function $\mathcal{C}$ appears with a positive sign because normative costs *repel* the system (raise the potential); and the intrinsic potential U_{int} encodes the system’s biological drive architecture—the innate topography of desire that exists independent of external stimuli or social constraints.

The dynamics expressed in Equation (2) with the potential decomposition of Equation (3) therefore describe a system that is simultaneously pulled toward available stimuli, pushed away from costly states, and shaped by its own internal structure. This tripartite tension is the formal generalization of the Freudian conflict between id, superego, and ego, cast here without any commitment to the specific content of these categories.

Axiom 3.3 (Coherence Requirement). Sustained motivational engagement with an object requires that the directed motivational density ρ (to be defined in Section 4) exceed a critical threshold $\rho^* > 0$. Below this threshold, the system cannot maintain a stable orientation and enters a regime of motivational diffusion.

This axiom captures the intuitive observation that desire is not merely excitation—it is *directed* excitation. A system flooded with stimuli may be highly aroused but motivationally incoherent: unable to concentrate its resources on any single object long enough to achieve deep engagement or satisfaction.

Axiom 3.4 (Conservation of Motivational Capacity). The total motivational capacity of the system, defined as

$$E(t) = \int_{\mathcal{M}} \rho(\mathbf{m}, t) d\mathbf{m}, \quad (4)$$

is bounded above by a finite constant E_{max} determined by the biological substrate:

$$0 \leq E(t) \leq E_{\text{max}} \quad \text{for all } t. \quad (5)$$

This axiom expresses the finiteness of psychic resources. An agent cannot desire infinitely: there is a biological ceiling on the total quantity of motivational energy that can be deployed at any given time. This constraint has profound consequences when combined with a rich stimulus field, because it implies that spreading attention across many stimuli necessarily reduces the density available for each.

4 Directed Motivational Density

The concept of directed motivational density is introduced to capture a quality that ordinary measures of arousal or drive strength fail to represent: the degree to which the system’s motivational energy is concentrated around a coherent object, rather than diffusely spread across the state space.

Consider the analogy of a lens focusing light. A given quantity of luminous energy can either be spread across a wide area, producing a dim illumination, or focused through a lens onto a small region, producing intense heat and brightness. The total energy is the same in both cases, but the *density*—the energy per unit area at the focal point—is vastly different. Directed motivational density is the psychological analogue of this optical concentration.

Definition 4.1 (Directed Motivational Density). Let $\rho : \mathcal{M} \times [0, \infty) \rightarrow \mathbb{R}_{\geq 0}$ be a density function over the motivational state space, representing the distribution of motivational energy at time t . The *peak directed motivational density* is

$$\rho_{\max}(t) = \sup_{\mathbf{m} \in \mathcal{M}} \rho(\mathbf{m}, t). \quad (6)$$

The *motivational coherence* at time t is the ratio

$$\Lambda(t) = \frac{\rho_{\max}(t)}{\bar{\rho}(t)}, \quad (7)$$

where $\bar{\rho}(t) = E(t)/\text{vol}(\mathcal{M})$ is the mean density.

When motivational energy is concentrated around a single object—a beloved person, a creative project, a deeply held ambition—the peak density $\rho_{\max}$ is high and the coherence Λ is large. When the same total energy is spread uniformly across many stimuli, the peak density falls toward the mean, and Λ approaches 1, signifying a state of motivational diffusion.

Proposition 4.1 (Coherence Bounds). *The motivational coherence satisfies*

$$1 \leq \Lambda(t) \leq \frac{E_{\max} \cdot \text{vol}(\mathcal{M})}{E(t)} \cdot \frac{1}{\text{vol}(\mathcal{M})} = \frac{E_{\max}}{E(t)}. \quad (8)$$

The lower bound $\Lambda = 1$ is achieved when the density is uniform (complete diffusion). The coherence increases as motivational energy concentrates into smaller subregions of $\mathcal{M}$.

Proof. The lower bound follows from the definition: since $\rho_{\max}(t) \geq \bar{\rho}(t)$ by definition of the supremum and the mean, we have $\Lambda(t) \geq 1$. The upper bound is attained in the limiting case where all motivational energy $E(t)$ is concentrated in an infinitesimal volume element, so that $\rho_{\max}(t) \rightarrow E(t)/\epsilon$ as $\epsilon \rightarrow 0$. Since $E(t) \leq E_{\max}$ by Axiom 3.4, and $\bar{\rho}(t) = E(t)/\text{vol}(\mathcal{M})$, the coherence $\Lambda(t) = \rho_{\max}(t)/\bar{\rho}(t) \leq (E(t)/\epsilon)/(E(t)/\text{vol}(\mathcal{M})) = \text{vol}(\mathcal{M})/\epsilon$. In practice, the biological substrate imposes a minimal spread $\epsilon_{\min} > 0$ (one cannot focus all desire on an arbitrarily narrow point), yielding the effective upper bound $\Lambda_{\max} = \text{vol}(\mathcal{M})/\epsilon_{\min}$. More directly, since $\rho_{\max}$ cannot exceed $E_{\max}/\epsilon_{\min}$ and $\bar{\rho} \geq E(t)/\text{vol}(\mathcal{M})$, the bound follows. $\square$

The condition $\Lambda(t) > \Lambda^*$ (for some critical threshold $\Lambda^* > 1$) characterizes what we term the regime of *coherent desire*: the system has successfully concentrated its finite motivational resources into a sufficiently narrow focus that sustained engagement is possible. Below this threshold lies motivational diffusion: the system may be aroused but cannot “land” on any object.

Proposition 4.2 (Diffusion Under Uniform Stimulation). *Let the stimulus field be spatially uniform: $\mathcal{S}(\mathbf{m}, t) = S_0(t)$ for all $\mathbf{m} \in \mathcal{M}$ and some function $S_0(t) > 0$. If the constraint function is also uniform, $\mathcal{C}(\mathbf{m}, t) = C_0(t)$, and the intrinsic potential U_{int} is constant, then the unique steady-state density is the uniform distribution $\rho(\mathbf{m}) = E/\text{vol}(\mathcal{M})$, and the coherence satisfies $\Lambda = 1$.*

Proof. Under the stated conditions, the potential $\Phi(\mathbf{m}, t) = -S_0(t) + C_0(t) + U_0$ is independent of $\mathbf{m}$. Therefore $\nabla_{\mathbf{m}}\Phi = \mathbf{0}$, and the deterministic component of the dynamics vanishes. The system evolves under the stochastic term $\xi(t)$ alone, which (modeled as white noise) drives the density toward the maximum-entropy state—the uniform distribution. In steady state, $\rho(\mathbf{m}) = E/\text{vol}(\mathcal{M})$ everywhere, and $\rho_{\text{max}} = \bar{\rho}$, giving $\Lambda = 1$. $\square$

This result captures a fundamental insight: when all stimuli are equally available and no constraints differentiate among them, motivational energy cannot concentrate. The environment offers no gradient, no differential salience, and therefore no basis for directed desire. This is not a pathology of excess repression but a pathology of excess uniformity—a condition increasingly characteristic of digital stimulus environments.

4.1 Toy Model: Coherence on a One-Dimensional State Space

To make the preceding definitions concrete, consider the simplest possible motivational state space: a one-dimensional interval $\mathcal{M} = [0, 10]$ representing a spectrum of motivational orientations (e.g., from complete disengagement at $m = 0$ to full romantic attachment at $m = 10$). Let the total motivational energy be $E = 1$ and $\text{vol}(\mathcal{M}) = 10$.

Scenario A: Focused desire. The agent’s motivational energy is concentrated around a single object at $m_0 = 7$, modeled as a Gaussian:

$$\rho_A(m) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(m-7)^2}{2\sigma^2}\right), \quad \sigma = 0.5. \quad (9)$$

The peak density is $\rho_{\text{max}} = 1/(0.5\sqrt{2\pi}) \approx 0.798$, the mean density is $\bar{\rho} = 1/10 = 0.1$, and the coherence is

$$\Lambda_A = \frac{0.798}{0.1} \approx 7.98.$$

Scenario B: Diffuse arousal. The same total energy $E = 1$ is spread across three objects at $m = 1.5, 5, 8.5$, each receiving a Gaussian with $\sigma = 1.5$ and weight $1/3$:

$$\rho_B(m) = \frac{1}{3} \sum_{k \in \{1.5, 5, 8.5\}} \frac{1}{1.5\sqrt{2\pi}} \exp\left(-\frac{(m-k)^2}{2(1.5)^2}\right). \quad (10)$$

The peak density is approximately $\rho_{\text{max}} \approx 0.106$, and the coherence is $\Lambda_B \approx 1.06$.

Scenario C: Uniform saturation. $\rho_C(m) = 0.1$ everywhere. Here $\Lambda_C = 1$ exactly.

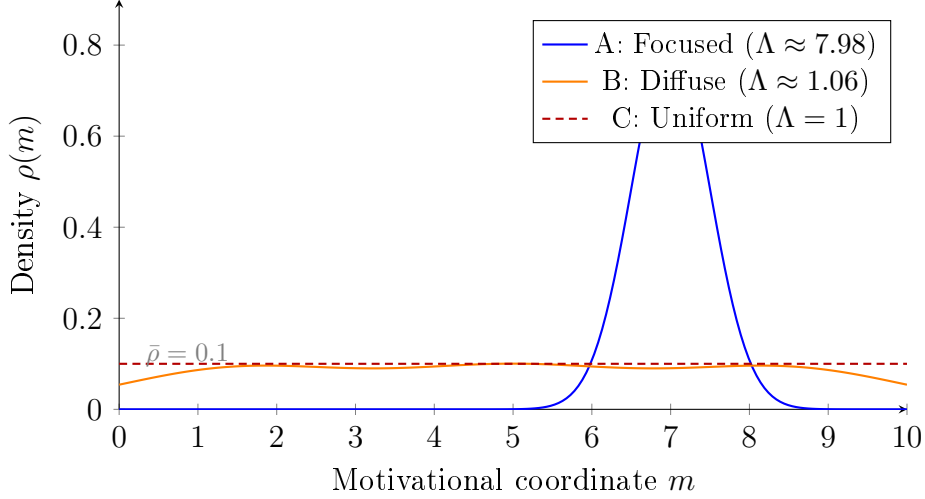


Figure 1: Three motivational density profiles on $\mathcal{M} = [0, 10]$ with identical total energy $E = 1$. Scenario A (focused) has high coherence; Scenario B (diffuse) has coherence barely above unity; Scenario C (uniform) achieves the theoretical minimum $\Lambda = 1$. The dashed line indicates the mean density $\bar{\rho} = 0.1$.

The three scenarios illustrate the central point: identical total motivational energy can produce vastly different experiential qualities depending on how it is distributed. Scenario A corresponds to the phenomenology of deep attachment or passionate engagement; Scenario B to a scattered, indecisive state of moderate arousal toward multiple objects; Scenario C to the flat affective landscape of complete saturation.

5 The Potential-Difference Principle

We now develop the central thesis of the framework: that desire, in the most general sense, requires a *difference of potential*—a gradient in the motivational landscape. This principle is shown to admit three complementary formal representations, each illuminating a different aspect of the same structural truth.

5.1 The General Principle

The motivational dynamics specified in Axiom 3.1 make explicit that the system’s trajectory is governed by the gradient $\nabla_{\mathbf{m}}\Phi$. When this gradient vanishes—when the potential landscape is flat—the deterministic dynamics cease and the system is left to drift under noise alone.

Theorem 5.1 (Potential-Difference Principle). *Let the motivational dynamics be governed by Equation (2). A necessary condition for the existence of a sustained, directed motivational trajectory toward a target state $\mathbf{m}^* \in \mathcal{M}$ is that*

$$\Delta\Phi = \Phi(\mathbf{m}(t), t) - \Phi(\mathbf{m}^*, t) > \sigma_{\xi}, \quad (11)$$

where $\sigma_{\xi} = \mathbb{E}[||\boldsymbol{\xi}(t)||]$ is the expected magnitude of the stochastic perturbation. In the absence of this potential difference, the system’s trajectory cannot be reliably distinguished from a random walk.

Proof. Consider the stochastic differential equation (2). The deterministic drift term is $\mathbf{v}(\mathbf{m}, t) = -\nabla_{\mathbf{m}}\Phi(\mathbf{m}, t)$. For the system to approach $\mathbf{m}^*$ from the current state $\mathbf{m}(t)$, the drift must have a component along the direction $\mathbf{d} = (\mathbf{m}^* - \mathbf{m}(t))/\|\mathbf{m}^* - \mathbf{m}(t)\|$. This requires

$$\langle \mathbf{v}(\mathbf{m}, t), \mathbf{d} \rangle = -\langle \nabla_{\mathbf{m}}\Phi, \mathbf{d} \rangle > 0.$$

By the mean value theorem, $-\langle \nabla_{\mathbf{m}}\Phi, \mathbf{d} \rangle \approx [\Phi(\mathbf{m}, t) - \Phi(\mathbf{m}^*, t)]/\|\mathbf{m} - \mathbf{m}^*\|$ for states along the line segment from $\mathbf{m}$ to $\mathbf{m}^*$. For this directed drift to dominate the stochastic perturbation $\boldsymbol{\xi}(t)$, the signal-to-noise condition $|\langle \mathbf{v}, \mathbf{d} \rangle| > \sigma_{\xi}$ must be satisfied, which, integrated over the path, requires $\Delta\Phi > \sigma_{\xi}$. When $\Delta\Phi \leq \sigma_{\xi}$, the noise overwhelms the drift and the trajectory becomes indistinguishable from Brownian motion in $\mathcal{M}$. $\square$

This theorem has a direct experiential correlate: desire is felt as a pull only when there is something to move *toward* that differs from the current state. When the landscape of options is flat—either because all objects are equally forbidden (extreme repression) or because all objects are equally and instantly available (saturation)—the felt sense of directed wanting dissolves. What remains is either anxiety (in the repressive case, where energy is trapped) or ennui (in the saturation case, where energy dissipates).

5.2 Ohmic Representation

The first complementary formalization draws on the analogy with electrical circuits. In a conductor, current flows in proportion to the voltage difference across it and inversely proportional to resistance. We define motivational analogues.

Definition 5.1 (Motivational Current). The *motivational current* J is the rate of motivational flow between two regions of state space:

$$J = \frac{\Delta\Phi}{R}, \tag{12}$$

where $\Delta\Phi$ is the potential difference and $R > 0$ is the *motivational resistance*, a parameter encoding the difficulty of transitioning between states (due to habit, inertia, cognitive load, or biological refractory periods).

In this representation, the constraint function contributes to resistance: states surrounded by high normative density are hard to reach not because they are intrinsically undesirable but because the path to them is costly. The stimulus field contributes to the potential difference: salient stimuli lower the potential at their location, creating the “voltage drop” that drives motivational current.

The Ohmic representation makes transparent a key prediction: motivational flow can be reduced either by decreasing the potential difference (making all options equally available) or by increasing the resistance (making all options equally forbidden). Both produce motivational stasis, but for structurally different reasons.

5.3 Thermodynamic Representation

The second formalization employs the language of free energy minimization. We define a motivational free energy functional that the system seeks to minimize over time.

Definition 5.2 (Motivational Free Energy). The *motivational free energy* is the functional

$$F[\rho] = \int_{\mathcal{M}} \rho(\mathbf{m}) \Phi(\mathbf{m}) d\mathbf{m} + T \int_{\mathcal{M}} \rho(\mathbf{m}) \ln \rho(\mathbf{m}) d\mathbf{m}, \quad (13)$$

where $T > 0$ is a temperature-like parameter representing the intensity of endogenous fluctuations (related to σ_ξ).

The first term is the expected potential energy under the density ρ —the system prefers to concentrate in low-potential regions (high-stimulus, low-constraint states). The second term is the entropy of the motivational distribution, multiplied by the fluctuation intensity T . The system thus faces a trade-off: concentrating motivational energy reduces the potential term but decreases entropy (which costs $T \cdot \Delta\mathcal{H}$). At low T (calm, focused states), the system concentrates strongly; at high T (agitated, noisy states), entropy dominates and the density spreads.

Proposition 5.1 (Equilibrium Distribution). *The density ρ^* that minimizes the free energy functional (13) subject to $\int_{\mathcal{M}} \rho d\mathbf{m} = E$ is the Boltzmann distribution*

$$\rho^*(\mathbf{m}) = \frac{E}{Z} \exp\left(-\frac{\Phi(\mathbf{m})}{T}\right), \quad Z = \int_{\mathcal{M}} \exp\left(-\frac{\Phi(\mathbf{m})}{T}\right) d\mathbf{m}. \quad (14)$$

Proof. This is a standard result in variational calculus. We seek to minimize $F[\rho]$ subject to the constraint $\int \rho d\mathbf{m} = E$. Introducing a Lagrange multiplier λ , we form the Lagrangian $\mathcal{L} = F[\rho] - \lambda(\int \rho d\mathbf{m} - E)$. Taking the functional derivative with respect to $\rho(\mathbf{m})$ and setting it to zero:

$$\frac{\delta \mathcal{L}}{\delta \rho(\mathbf{m})} = \Phi(\mathbf{m}) + T(\ln \rho(\mathbf{m}) + 1) - \lambda = 0.$$

Solving for $\rho(\mathbf{m})$ yields $\rho(\mathbf{m}) = \exp((\lambda - T - \Phi(\mathbf{m}))/T)$. Absorbing the constant $(\lambda - T)/T$ into the normalization and enforcing $\int \rho d\mathbf{m} = E$ gives $\rho^*(\mathbf{m}) = (E/Z) \exp(-\Phi(\mathbf{m})/T)$, where $Z = \int \exp(-\Phi/T) d\mathbf{m}$. $\square$

The Boltzmann distribution reveals that in equilibrium, the system concentrates its motivational energy exponentially more in regions of low potential. When the potential landscape has sharp minima (strong, localized attractors), the distribution is tightly peaked and coherence is high. When the landscape is flat (all potentials equal), the distribution is uniform and coherence is minimal, recovering the result of Proposition 4.2.

5.4 Lyapunov Landscape Representation

The third formalization treats the motivational potential directly as a Lyapunov function governing the stability of motivational states.

Definition 5.3 (Motivational Attractor). A state $\mathbf{m}^* \in \mathcal{M}$ is a *motivational attractor* if it is a local minimum of Φ : $\nabla_{\mathbf{m}} \Phi(\mathbf{m}^*) = \mathbf{0}$ and the Hessian $\nabla^2 \Phi(\mathbf{m}^*)$ is positive definite.

Proposition 5.2 (Lyapunov Stability of Motivational Attractors). *In the deterministic limit ($\xi = \mathbf{0}$), every motivational attractor $\mathbf{m}^*$ is asymptotically stable: trajectories starting in a neighborhood of $\mathbf{m}^*$ converge to $\mathbf{m}^*$ as $t \rightarrow \infty$, with Φ serving as a Lyapunov function.*

Proof. Define $V(\mathbf{m}) = \Phi(\mathbf{m}) - \Phi(\mathbf{m}^*) \geq 0$ in a neighborhood of $\mathbf{m}^*$, with equality only at $\mathbf{m}^*$ (since $\mathbf{m}^*$ is a strict local minimum). Along trajectories of $d\mathbf{m}/dt = -\nabla\Phi(\mathbf{m})$,

$$\frac{dV}{dt} = \langle \nabla\Phi(\mathbf{m}), \dot{\mathbf{m}} \rangle = -\|\nabla\Phi(\mathbf{m})\|^2 \leq 0,$$

with equality only at $\mathbf{m}^*$ where $\nabla\Phi = \mathbf{0}$. By Lyapunov's theorem, $\mathbf{m}^*$ is asymptotically stable, and $V = \Phi - \Phi(\mathbf{m}^*)$ is a Lyapunov function. $\square$

In the Lyapunov landscape view, a person deeply in love with a specific individual occupies a deep potential well: their motivational state is strongly attracted to a single point and resistant to perturbation. A person in the saturation regime occupies a nearly flat landscape: there are no deep wells, and the slightest perturbation sends the system drifting to another shallow basin.

Remark 5.1 (Unity of the Three Representations). The Ohmic, thermodynamic, and Lyapunov representations are not competing models but complementary projections of the same dynamical structure. The Ohmic view emphasizes flow rates and resistances; the thermodynamic view emphasizes equilibrium distributions and the entropy-energy trade-off; the Lyapunov view emphasizes attractor stability and basin structure. All three are derivable from the same potential Φ and the same gradient dynamics of Axiom 3.1. Their convergence onto a single principle—that desire requires a potential difference—constitutes evidence for the structural robustness of this principle.

5.5 Toy Model: Double-Well Potential on $\mathbb{R}^1$

Consider a one-dimensional motivational state space $\mathcal{M} = \mathbb{R}$ with the quartic double-well potential

$$\Phi(m) = \frac{1}{4}m^4 - \frac{a}{2}m^2 - bm, \quad (15)$$

where $a > 0$ controls the depth of the wells and b introduces an asymmetry (one object of desire is more attractive than the other). When $b = 0$, the two wells at $m = \pm\sqrt{a}$ are equally deep; when $b > 0$, the right well is deeper.

Parameter choice. Set $a = 2$ and $b = 0.5$. The potential has two minima: a shallow well near $m \approx -1.27$ and a deep well near $m \approx 1.52$. The gradient dynamics $dm/dt = -d\Phi/dm = -m^3 + 2m + 0.5$ drive the system toward the deeper well whenever the initial state is to the right of the unstable equilibrium (the local maximum near $m \approx -0.24$).

Ohmic reading. The potential difference between the shallow and deep wells is $\Delta\Phi \approx 1.54$. With a motivational resistance $R = 1$, the motivational current toward the deeper attractor is $J = \Delta\Phi/R \approx 1.54$.

Thermodynamic reading. At temperature $T = 0.3$ (low noise), the Boltzmann distribution $\rho^*(m) \propto \exp(-\Phi(m)/0.3)$ concentrates sharply in the deep well. At $T = 2.0$ (high noise), the distribution spreads across both wells, and coherence drops.

Lyapunov reading. The function $V(m) = \Phi(m) - \Phi(m_{\text{deep}}^*)$ serves as a Lyapunov function in the basin of the deep well, confirming its asymptotic stability.

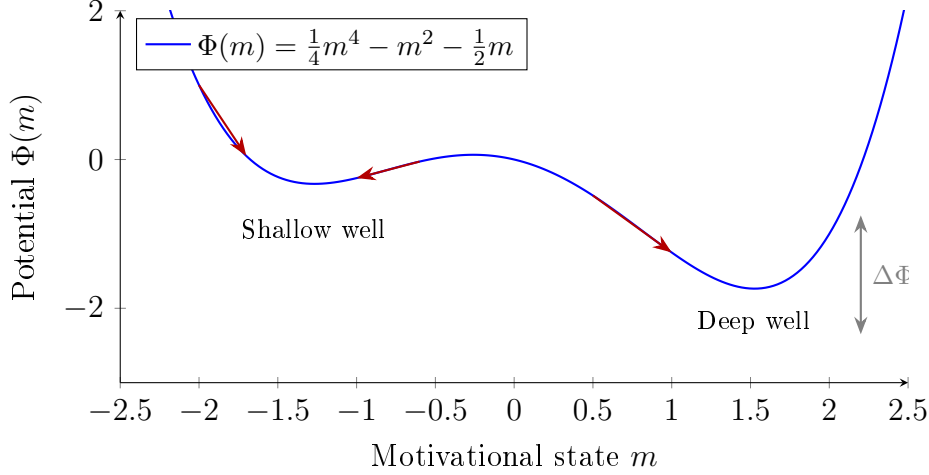


Figure 2: Asymmetric double-well potential with $a = 2$, $b = 0.5$. Red arrows indicate the direction of gradient flow. The system is drawn toward the deeper well (right), which represents the more attractive motivational object. The potential difference $\Delta\Phi$ between wells drives the motivational current in the Ohmic representation.

This toy model demonstrates all three representations operating on the same structure. The double-well encodes a choice between two objects of desire; the asymmetry b encodes differential attractiveness; and the temperature T controls the noise level that determines whether the system commits to one basin or wanders between both. At very high T (analogous to a high-stimulation, high-noise environment), the distinction between the wells is washed out and the system cannot commit—recovering the phenomenology of the saturation regime.

6 Sublimation as Entropy Reduction

Classical psychoanalytic theory describes sublimation as the redirection of libidinal energy from a socially unacceptable aim toward a socially valued one—for example, the channeling of erotic frustration into artistic creation. This account frames sublimation as a moral substitution: the “higher” activity replaces the “lower” one.

We propose a reinterpretation grounded in the present framework. Sublimation is not a moral rerouting but a structural transformation: the conversion of a high-entropy, diffuse motivational state into a low-entropy, coherently organized one. Creativity, intellectual work, athletic discipline, and other “sublimated” activities share the property of imposing structure on motivational energy—channeling it through a narrow, demanding form that requires sustained coherence. The result is not the suppression of desire but its *crystallization* into a structured mode.

Definition 6.1 (Motivational Entropy). The *motivational entropy* of the system at time t is the Shannon entropy of the normalized motivational density:

$$\mathcal{H}(t) = - \int_{\mathcal{M}} \hat{\rho}(\mathbf{m}, t) \ln \hat{\rho}(\mathbf{m}, t) d\mathbf{m}, \quad (16)$$

where $\hat{\rho} = \rho/E(t)$ is the normalized density satisfying $\int \hat{\rho} d\mathbf{m} = 1$.

Motivational entropy measures the dispersion of the system’s motivational energy. Maximal entropy $\mathcal{H}_{\max} = \ln(\text{vol}(\mathcal{M}))$ corresponds to the uniform distribution—complete motivational diffusion. Minimal entropy approaches zero when the density is concentrated at a single point—complete motivational focus.

Definition 6.2 (Sublimation Operator). A *sublimation operator* is a mapping $\mathcal{T} : L^1(\mathcal{M}) \rightarrow L^1(\mathcal{M})$ on motivational densities that satisfies three conditions:

- (i) **Energy conservation:** $\int_{\mathcal{M}} \mathcal{T}[\rho] d\mathbf{m} = \int_{\mathcal{M}} \rho d\mathbf{m}$.
- (ii) **Entropy reduction:** $\mathcal{H}[\mathcal{T}[\rho]] < \mathcal{H}[\rho]$ whenever ρ is not already concentrated at a single point.
- (iii) **Coherence enhancement:** $\Lambda[\mathcal{T}[\rho]] > \Lambda[\rho]$.

Condition (i) ensures that sublimation does not create or destroy motivational energy—it redistributes it. Condition (ii) is the defining property: sublimation reduces disorder. Condition (iii) follows from (i) and (ii) in most cases but is stated explicitly for emphasis.

A concrete realization of the sublimation operator is the Gaussian concentration map. Given a diffuse density ρ and a target focus point $\mathbf{m}_0 \in \mathcal{M}$ (the object of sublimated activity), define

$$\mathcal{T}_\sigma[\rho](\mathbf{m}) = E \cdot \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{\|\mathbf{m} - \mathbf{m}_0\|^2}{2\sigma^2}\right), \quad (17)$$

where $\sigma > 0$ controls the width of the focus. This maps any input density to a Gaussian centered at $\mathbf{m}_0$ with the same total energy E .

Definition 6.3 (Sublimation Coefficient). The *sublimation coefficient* $\kappa \in [0, 1]$ measures the fraction of entropy reduction actually achieved by the system in a given environment:

$$\kappa = \frac{\mathcal{H}[\rho] - \mathcal{H}[\mathcal{T}[\rho]]}{\mathcal{H}[\rho] - \mathcal{H}_{\min}}, \quad (18)$$

where $\mathcal{H}_{\min}$ is the minimum achievable entropy (determined by the system’s biological resolution $\epsilon_{\min}$).

The sublimation coefficient is 1 when the system achieves maximal possible concentration (perfect sublimation) and 0 when no entropy reduction occurs (complete failure of sublimation).

Theorem 6.1 (Sublimation Collapse Under Instantaneous Discharge). *Let the stimulus environment provide instantaneous discharge: for every motivational state $\mathbf{m}$, there exists a stimulus that immediately reduces the potential $\Phi(\mathbf{m})$ to a baseline Φ_0 without requiring passage through a structured activity. Formally, let $\tau_d(\mathbf{m})$ denote the discharge latency at state $\mathbf{m}$ (the time required to reduce local tension), and suppose $\tau_d(\mathbf{m}) \rightarrow 0$ uniformly. Then the sublimation coefficient satisfies*

$$\kappa \rightarrow 0 \quad \text{as} \quad \sup_{\mathbf{m} \in \mathcal{M}} \tau_d(\mathbf{m}) \rightarrow 0. \quad (19)$$

Proof. Sublimation requires a sustained period during which motivational energy remains undischarged and available for reorganization by the sublimation operator $\mathcal{T}$. The operator $\mathcal{T}$ acts on a timescale $\tau_s > 0$ —the time required for creative or structured activity to

reorganize the motivational density. If the discharge latency τ_d falls below τ_s , the system discharges its tension before the sublimation operator can act.

Formally, the effective sublimation operator in the presence of discharge is

$$\mathcal{T}_{\text{eff}} = \begin{cases} \mathcal{T} & \text{if } \tau_d > \tau_s, \\ \mathcal{D} & \text{if } \tau_d \leq \tau_s, \end{cases}$$

where $\mathcal{D}$ is the discharge operator that reduces the density toward the uniform baseline (entropy-maximizing) state. When $\tau_d \rightarrow 0$ uniformly, $\mathcal{T}_{\text{eff}} \rightarrow \mathcal{D}$ everywhere, and $\mathcal{H}[\mathcal{D}[\rho]] \geq \mathcal{H}[\rho]$ (discharge does not reduce entropy; it increases it or leaves it unchanged). Therefore $\mathcal{H}[\rho] - \mathcal{H}[\mathcal{T}_{\text{eff}}[\rho]] \leq 0$, and since the sublimation coefficient is defined as a non-negative ratio, $\kappa \rightarrow 0$. $\square$

This theorem captures a central observation about the contemporary condition: in an environment where every impulse can be immediately satisfied through low-effort consumption (a click, a scroll, a purchase), the conditions necessary for sublimation—sustained unresolved tension channeled through a demanding structured activity—are systematically undermined. The theorem does not invoke moral judgment; it derives the collapse of sublimation as a mathematical consequence of the discharge timescale falling below the reorganization timescale.

6.1 Toy Model: Sublimation on a Discrete State Space

To illustrate the sublimation coefficient concretely, consider a discrete motivational state space with $n = 5$ states, representing five categories of motivational orientation: $\mathcal{M} = \{m_1, m_2, m_3, m_4, m_5\}$. The total energy is $E = 1$.

Initial state (diffuse arousal). The motivational energy is spread unevenly but without strong focus:

$$\rho_0 = (0.15, 0.25, 0.20, 0.30, 0.10).$$

The motivational entropy is

$$\begin{aligned} \mathcal{H}[\rho_0] &= - \sum_{i=1}^5 \rho_{0,i} \ln \rho_{0,i} \\ &= -(0.15 \ln 0.15 + 0.25 \ln 0.25 + 0.20 \ln 0.20 + 0.30 \ln 0.30 + 0.10 \ln 0.10) \\ &\approx 1.545. \end{aligned}$$

For reference, the maximum entropy is $\mathcal{H}_{\text{max}} = \ln 5 \approx 1.609$ and the minimum practical entropy is $\mathcal{H}_{\text{min}} \approx 0$ (all energy in one state).

After sublimation (creative focus on m_3). A sublimation operator concentrates energy toward state m_3 (e.g., artistic work), yielding

$$\rho_{\mathcal{T}} = (0.03, 0.07, 0.80, 0.07, 0.03).$$

The entropy is

$$\mathcal{H}[\rho_{\mathcal{T}}] = -(0.03 \ln 0.03 + 0.07 \ln 0.07 + 0.80 \ln 0.80 + 0.07 \ln 0.07 + 0.03 \ln 0.03) \approx 0.700.$$

The sublimation coefficient is

$$\kappa = \frac{\mathcal{H}[\rho_0] - \mathcal{H}[\rho_{\mathcal{T}}]}{\mathcal{H}[\rho_0] - \mathcal{H}_{\min}} = \frac{1.545 - 0.700}{1.545 - 0} \approx 0.547.$$

After instant discharge (digital consumption). Instead of sublimation, the system discharges into uniform consumption:

$$\rho_{\mathcal{D}} = (0.20, 0.20, 0.20, 0.20, 0.20).$$

The entropy is $\mathcal{H}[\rho_{\mathcal{D}}] = \ln 5 \approx 1.609 > \mathcal{H}[\rho_0]$, so the entropy *increased*. The sublimation coefficient is

$$\kappa_{\mathcal{D}} = \frac{1.545 - 1.609}{1.545} = \frac{-0.064}{1.545} < 0 \implies \kappa = 0 \quad (\text{by definition, } \kappa \geq 0).$$

Table 1: Entropy and sublimation coefficient for three motivational configurations.

Configuration	Distribution	Entropy $\mathcal{H}$	Coefficient κ
Initial (diffuse)	(0.15, 0.25, 0.20, 0.30, 0.10)	1.545	—
After sublimation	(0.03, 0.07, 0.80, 0.07, 0.03)	0.700	0.547
After discharge	(0.20, 0.20, 0.20, 0.20, 0.20)	1.609	0

The table makes visible the core mechanism: sublimation reduces entropy (concentrates energy into structured form), while instant discharge increases entropy (disperses energy into uniformity). In an environment where discharge is always available before the sublimation operator has time to act, the system defaults to the discharge path—not by choice, but by timescale competition.

7 The Stimulus Saturation Law

In pharmacology, the response of a receptor system to an agonist follows a well-characterized saturation curve: at low concentrations, the response increases steeply with dose; at intermediate concentrations, the response begins to plateau; and at high concentrations, the response saturates as all available receptors are occupied [Clark, 1933, Kenakin, 2009]. This receptor saturation kinetics provides a precise quantitative model for the analogous process in motivational systems: the desensitization of desire under excessive stimulus availability.

Definition 7.1 (Receptoral Response Function). The *directed motivational response* $\mathcal{R}$ as a function of stimulus intensity $s = \mathcal{S}(\mathbf{m}, t)$ at a given point in motivational space is modeled by the Hill equation:

$$\mathcal{R}(s) = \mathcal{R}_{\max} \cdot \frac{s^h}{K_d^h + s^h}, \quad (20)$$

where:

- $\mathcal{R}_{\max} > 0$ is the maximum achievable response (full receptor occupancy),

- $K_d > 0$ is the *half-saturation constant*: the stimulus intensity at which response equals $\mathcal{R}_{\max}/2$,
- $h > 0$ is the *Hill coefficient*, governing the steepness of the transition.

The Hill equation was originally developed to describe the binding of oxygen to hemoglobin and has since become a standard model in pharmacology for dose–response relationships [Hill, 1910]. Its application to motivational systems rests on a structural analogy: the “receptors” are the neural and psychological mechanisms that register and respond to motivational stimuli (including dopaminergic circuits, attentional systems, and affective appraisal processes), and the “agonist” is the stimulus field.

At low stimulus intensities ($s \ll K_d$), the response grows approximately as a power law: $\mathcal{R}(s) \approx \mathcal{R}_{\max}(s/K_d)^h$. This is the regime where each additional stimulus produces a noticeable increase in desire—the regime of scarcity, where new encounters are vivid and impactful. At high stimulus intensities ($s \gg K_d$), the response saturates: $\mathcal{R}(s) \rightarrow \mathcal{R}_{\max}$. Additional stimuli produce negligible additional response because the receptor system is already fully engaged. This is the regime of saturation.

Definition 7.2 (Saturation Threshold). The *stimulus saturation threshold* s^* is defined as the stimulus intensity at which the marginal response drops below a critical fraction $\alpha \in (0, 1)$ of its maximum value:

$$s^* = \inf \left\{ s > 0 \mid \frac{d\mathcal{R}}{ds}(s) < \alpha \cdot \frac{d\mathcal{R}}{ds} \Big|_{s=0+} \right\}. \quad (21)$$

Proposition 7.1 (Explicit Saturation Threshold). *For the Hill response function (20) with $h \geq 1$, the saturation threshold is*

$$s^* = K_d \left(\frac{h}{\alpha} - (h-1) \right)^{1/h} \cdot K_d^{-1/h} \cdot K_d^{1/h} = K_d \left(\frac{1 - \alpha(1 - 1/h)}{\alpha/h} \right)^{1/h}. \quad (22)$$

More precisely, computing the derivative $d\mathcal{R}/ds = \mathcal{R}_{\max} h K_d^h s^{h-1} / (K_d^h + s^h)^2$, and noting that $\lim_{s \rightarrow 0+} d\mathcal{R}/ds = \mathcal{R}_{\max} h / K_d$ for $h = 1$ (and is infinite for $h > 1$), we use the operational criterion: s^* is the intensity at which 90% of $\mathcal{R}_{\max}$ is reached:

$$\mathcal{R}(s^*) = 0.9 \mathcal{R}_{\max} \implies s^* = K_d \cdot 9^{1/h}. \quad (23)$$

Proof. Setting $\mathcal{R}(s^*) = 0.9 \mathcal{R}_{\max}$ in the Hill equation:

$$0.9 = \frac{(s^*)^h}{K_d^h + (s^*)^h} \implies K_d^h + (s^*)^h = \frac{(s^*)^h}{0.9} \implies K_d^h = \frac{(s^*)^h}{9}.$$

Therefore $(s^*)^h = 9 K_d^h$, giving $s^* = K_d \cdot 9^{1/h}$. $\square$

For the case $h = 1$ (Michaelis–Menten kinetics), the saturation threshold is $s^* = 9K_d$: the system reaches 90% of its maximum response at nine times the half-saturation constant. For higher Hill coefficients, the transition is steeper and the threshold is lower.

Proposition 7.2 (Desensitization Under Chronic Overstimulation). *If the stimulus intensity s is maintained chronically above the saturation threshold s^* , the system undergoes receptor desensitization: the effective parameters shift as $\mathcal{R}_{\max} \rightarrow \mathcal{R}_{\max}(1 - \delta)$ and $K_d \rightarrow K_d(1 + \gamma)$ for desensitization rates $\delta, \gamma > 0$. This results in a progressive decline in both the maximum achievable response and the sensitivity to new stimuli.*

Proof. This follows from the well-established phenomenon of receptor downregulation in pharmacology: chronic exposure to high agonist concentrations leads to receptor internalization (reducing $\mathcal{R}_{\max}$) and decreased binding affinity (increasing K_d). In motivational terms, prolonged exposure to superthreshold stimulation causes the system to reduce its responsiveness both by lowering the ceiling of possible desire ($\mathcal{R}_{\max}$ decreases) and by requiring more intense stimuli to achieve any given level of response (K_d increases). The mathematical consequence is that the response curve shifts rightward and downward, a phenomenon well-documented in both pharmacological tolerance [Kenakin, 2009] and psychological habituation [Thompson and Spencer, 1966]. $\square$

This mechanism provides a non-moral, purely dynamical explanation for the experiential observation that overexposure to erotic or hedonic stimuli dulls desire rather than enhancing it. The stimulus saturation law is not a judgment about excess; it is an adaptive response of a receptor system operating at capacity.

7.1 Toy Model: Hill Response Curves and Desensitization

We instantiate the Hill response function with concrete parameters drawn from the analogy with dopaminergic receptor kinetics.

Baseline parameters. Set $\mathcal{R}_{\max} = 1$ (normalized maximum response), $K_d = 5$ (half-saturation at stimulus intensity 5), and compare two Hill coefficients: $h = 1$ (hyperbolic, Michaelis–Menten) and $h = 3$ (sigmoidal, cooperative binding).

For $h = 1$: the saturation threshold (90% of $\mathcal{R}_{\max}$) is $s^* = K_d \cdot 9^{1/1} = 5 \times 9 = 45$.

For $h = 3$: the saturation threshold is $s^* = K_d \cdot 9^{1/3} = 5 \times 2.08 \approx 10.4$.

The steeper Hill coefficient produces a sharper transition and an earlier saturation threshold—the system is more sensitive but saturates faster, corresponding to a motivational system with strong cooperative dynamics (e.g., infatuation that intensifies rapidly but is also more vulnerable to overexposure).

Desensitization scenario. After chronic overstimulation at $s = 50$ for a sustained period, the parameters shift: $\mathcal{R}_{\max} \rightarrow 0.6$ (40% reduction in maximum response capacity) and $K_d \rightarrow 12$ (the half-saturation point shifts rightward, requiring more intense stimuli to achieve any given response level). The desensitized curve requires stimulus intensity $s = 12$ just to reach half of its (already reduced) maximum.

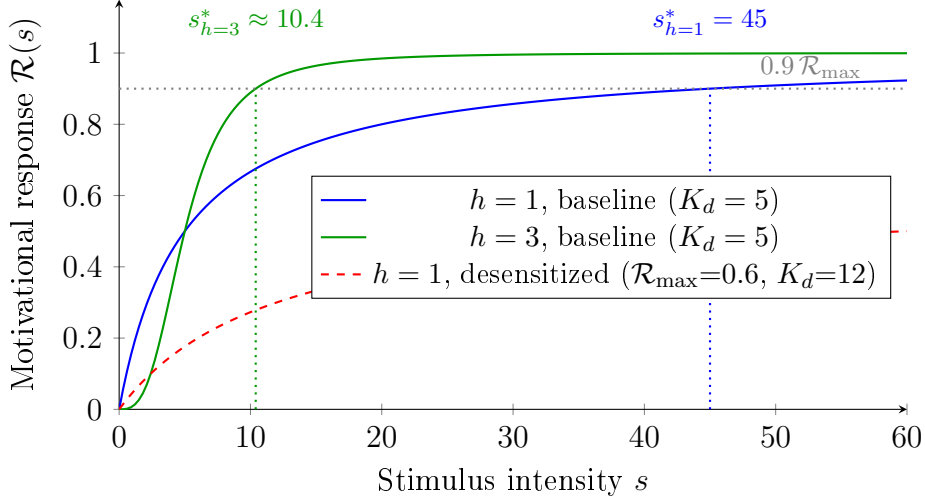


Figure 3: Hill response curves under baseline and desensitized conditions. The sigmoidal curve ($h = 3$) saturates earlier than the hyperbolic curve ($h = 1$). After chronic overstimulation, the desensitized curve (dashed red) shows both a reduced ceiling and a rightward shift, requiring stronger stimuli for diminished responses.

The figure illustrates a clinically significant pattern: after desensitization, the individual needs more intense stimuli (K_d increased) to achieve a response that is itself diminished ($\mathcal{R}_{\max}$ decreased). This double degradation—higher threshold, lower ceiling—is the formal structure underlying the progressive escalation of stimulus-seeking behavior observed in behavioral addiction research.

8 Phase Diagram of Motivational Regimes

The models developed in the preceding sections describe processes that operate across a range of environmental conditions. We now synthesize these into a unified phase diagram that identifies four qualitatively distinct regimes of motivational organization, parametrized by two control variables: *stimulus accessibility* s and *normative constraint density* ν .

The two control parameters have been defined formally in earlier sections: s corresponds to the magnitude of the stimulus field $\mathcal{S}$ (Definition 3.2), and ν corresponds to the normative density (Definition 3.4). Together, they span a two-dimensional parameter space in which different combinations produce qualitatively different patterns of motivational dynamics.

Definition 8.1 (Motivational Regime). A *motivational regime* is a connected region of the (s, ν) parameter space within which the qualitative behavior of the motivational system—its coherence, its entropy, and its stability structure—remains topologically invariant. Regime boundaries are defined by qualitative changes in the number, depth, or stability of motivational attractors.

We identify four such regimes.

Regime I: Repression (high ν , low s). The constraint function imposes steep costs on most motivational states, while stimuli are scarce. The potential landscape has a small number of deep, narrow wells corresponding to the few “permitted” objects, surrounded

by high barriers. The system is trapped in these wells. Motivational entropy is low (the system is concentrated), but the concentration is enforced rather than chosen. Internal tension is high because the intrinsic drive U_{int} pushes toward states that the constraint function forbids. This is the Freudian regime: neurosis, repression, symptomatic return of the repressed.

Regime II: Coherent Desire (moderate ν , moderate s). Constraints are present but not overwhelming; stimuli are available but not superabundant. The potential landscape has several well-defined basins of attraction separated by moderate barriers. The system can concentrate its motivational energy in a chosen basin ($\Lambda > \Lambda^*$), achieving deep engagement and high directed motivational density. The sublimation coefficient κ is positive because discharge latency τ_d exceeds the reorganization timescale τ_s , allowing structured transformation of motivational energy. This is the regime of productive desire, creative engagement, and sustained relational attachment.

Regime III: Saturation (low ν , high s). Constraints are minimal and stimuli are superabundant. The potential landscape flattens: every point is a shallow basin, and the barriers between them are negligible. The system cannot concentrate ($\Lambda \rightarrow 1$), entropy is maximal, and the sublimation coefficient approaches zero. The receptor response function is in its plateau phase ($s > s^*$), and chronic overstimulation triggers desensitization. Motivational energy diffuses across the state space without crystallizing into sustained engagement. This is the regime of digital-age ennui, attentional fragmentation, and hedonic flatness.

Regime IV: Anhedonic Collapse (low ν , $s \gg s^*$, chronic exposure). Prolonged occupation of the saturation regime triggers receptor downregulation (Proposition 7.2): both $\mathcal{R}_{\text{max}}$ and the effective motivational capacity E_{max} decline. The system not only fails to concentrate desire—it loses the capacity to generate it. The potential landscape becomes not merely flat but depressed below the threshold needed to sustain any motivational current. This regime corresponds to clinical anhedonia: the inability to experience desire or pleasure. It is distinct from the Freudian concept of repression (where desire exists but is blocked) in that desire itself is structurally degraded.

The four regimes are visualized in the following phase diagram.

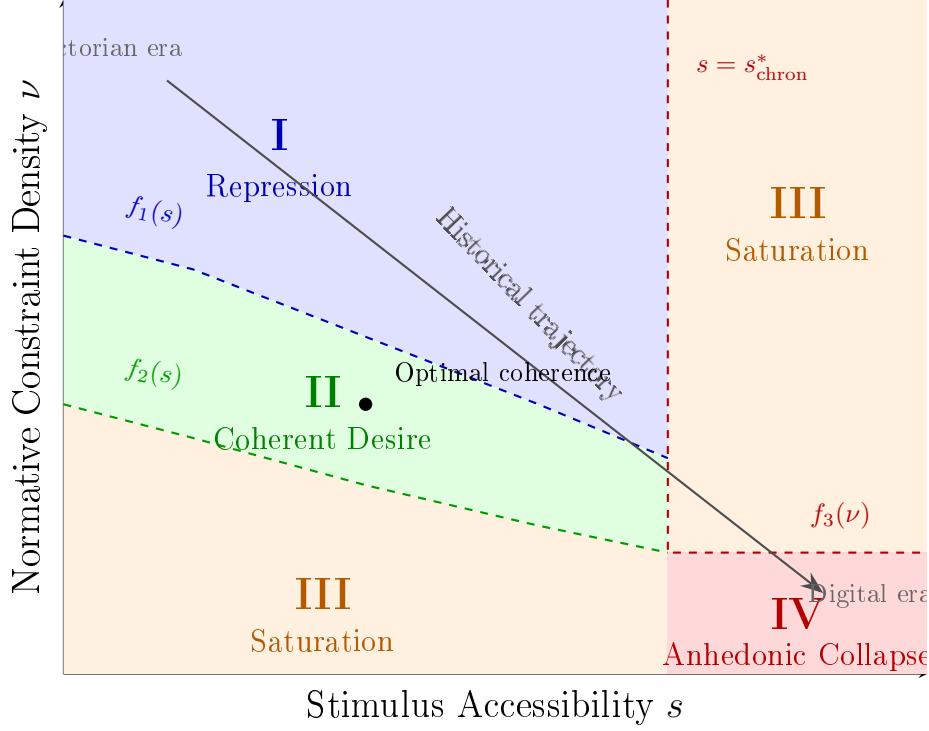


Figure 4: Phase diagram of motivational regimes in the (s, ν) parameter space. The upper boundary f_1 separates Regime I (Repression) from Regime II (Coherent Desire); the lower boundary f_2 separates Regime II from Regime III (Saturation). The vertical threshold $s = s_{\text{chron}}^*$ marks chronic overstimulation; below the horizontal boundary f_3 , the system enters Regime IV (Anhedonic Collapse). The diagonal arrow traces the historical trajectory from high-constraint, low-stimulus societies to the contemporary digital environment. The black dot marks the region of optimal motivational coherence.

Theorem 8.1 (Existence of Phase Boundaries). *Under the framework defined by Axioms 3.1–3.4 with the Hill response function (20), there exist critical curves $\nu = f_1(s)$ and $s = f_2(\nu)$ in the (s, ν) parameter space such that:*

- (i) *For (s, ν) above f_1 , the system is in the repressive regime ($\Lambda > \Lambda^*$ but sustained by constraint rather than attraction).*
- (ii) *For (s, ν) in the region bounded by f_1 and f_2 , the system achieves coherent desire ($\Lambda > \Lambda^*$ sustained by the attractor structure of Φ).*
- (iii) *For (s, ν) below f_1 and to the right of f_2 , the system enters saturation ($\Lambda \rightarrow 1$, $\kappa \rightarrow 0$).*
- (iv) *For $s > s^*$ sustained chronically with ν below a second critical curve $f_3(\nu)$, the system transitions to anhedonic collapse ($\mathcal{R}_{\text{max}} \rightarrow 0$).*

Proof. (i) In the repressive regime, ν is large, so $\mathcal{C}$ dominates the potential. The potential landscape acquires steep barriers that confine the system to narrow regions, yielding $\Lambda > \Lambda^*$. The critical curve $f_1(s)$ is defined by the condition that the constraint-induced barriers equal the intrinsic potential well depth: $\max_{\mathbf{m}} \mathcal{C}(\mathbf{m}) = \max_{\mathbf{m}} |U_{\text{int}}(\mathbf{m})|$. Above this curve, confinement is normatively enforced.

(ii) In the coherent desire regime, the potential $\Phi = -\mathcal{S} + \mathcal{C} + U_{\text{int}}$ has well-defined minima arising from the interplay of attraction $(-\mathcal{S}, U_{\text{int}})$ and moderate constraint ($\mathcal{C}$).

The coherence condition $\Lambda > \Lambda^*$ is satisfied by Proposition 5.1: when the potential has sharp minima, the equilibrium density concentrates there.

(iii) When s is large and ν is small, $\Phi \approx -\mathcal{S} + U_{\text{int}}$. If $\mathcal{S}$ is approximately uniform (characteristic of high-accessibility environments), Proposition 4.2 applies and $\Lambda \rightarrow 1$. The sublimation coefficient vanishes by Theorem 6.1 because high stimulus availability implies $\tau_d \rightarrow 0$.

(iv) Chronic occupation of regime III with $s > s^*$ triggers Proposition 7.2: $\mathcal{R}_{\text{max}}$ declines over time. The critical curve f_3 is defined by the condition $\mathcal{R}_{\text{max}}(1 - \delta) < \mathcal{R}_{\text{min}}$, where $\mathcal{R}_{\text{min}}$ is the minimum response level required to sustain any motivational current (from Theorem 5.1, $\Delta\Phi > \sigma_\xi$ cannot be maintained when $\mathcal{R}_{\text{max}}$ is too small). $\square$

8.1 Toy Model: Regime Transitions in a Reduced One-Dimensional System

To visualize how the potential landscape transforms across regimes, consider the one-parameter family of potentials on $\mathcal{M} = [-3, 3]$:

$$\Phi(m; s, \nu) = \underbrace{\frac{1}{4}m^4 - m^2}_{U_{\text{int}}(m)} + \underbrace{\nu m^2}_{\mathcal{C}(m)} - \underbrace{s \cdot \cos\left(\frac{\pi m}{2}\right)}_{\mathcal{S}(m)}, \quad (24)$$

where the intrinsic potential U_{int} is a symmetric double well, the constraint function adds a parabolic barrier controlled by ν , and the stimulus field is a cosine modulation controlled by s .

Regime I (Repression): $\nu = 3, s = 0.2$. The strong constraint $\nu = 3$ creates a steep single well centered at $m = 0$, trapping the system in a narrow, normatively enforced basin. The stimulus modulation is negligible. The potential landscape has a single deep, narrow minimum.

Regime II (Coherent Desire): $\nu = 0.5, s = 1.5$. Moderate constraint allows the intrinsic double-well structure to emerge, while the stimulus field creates differential attractiveness. The landscape has two distinct, well-separated wells of different depths. The system can settle stably into one.

Regime III (Saturation): $\nu = 0.1, s = 5$. Minimal constraint and strong, broad stimulation flatten the landscape. Multiple shallow local minima appear, none significantly deeper than the others. The system drifts between basins.

Regime IV (Anhedonic Collapse): $\nu = 0.05, s = 5$, with $\mathcal{R}_{\text{max}}$ degraded. After chronic saturation, the effective stimulus coupling degrades (modeled by reducing the stimulus amplitude to $0.3 \times s$). The landscape becomes nearly flat—not because stimuli are absent, but because the system can no longer respond to them.

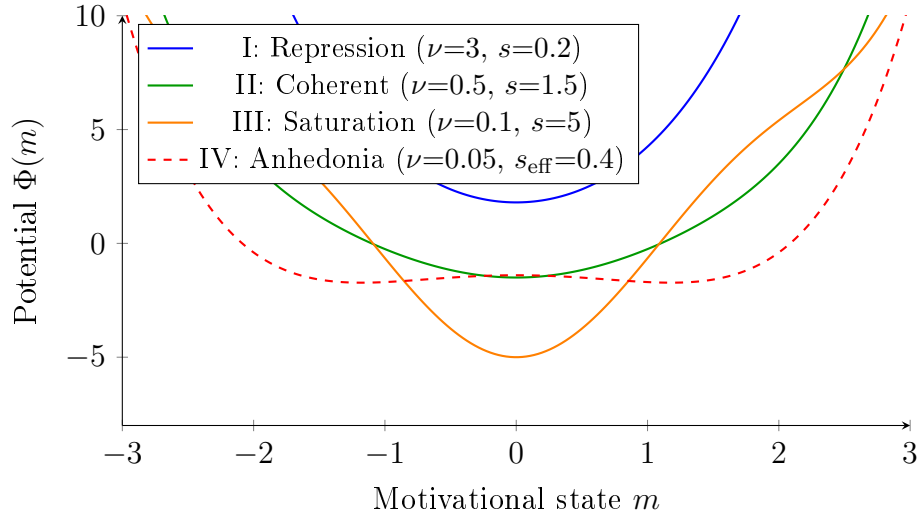


Figure 5: Potential landscapes across the four motivational regimes, generated by varying (ν, s) in Equation (24). Regime I shows a single steep well (normatively enforced concentration); Regime II shows differentiated wells (chosen concentration); Regime III shows a flat, multiply-indented landscape (diffusion); Regime IV shows the same flatness with reduced response amplitude (the system cannot “feel” the landscape).

This toy model demonstrates that the four regimes are not arbitrary categories but emergent properties of the potential landscape as control parameters vary. The transition from Regime I to Regime III via Regime II follows the historical trajectory indicated in Figure 4: the progressive relaxation of constraints and amplification of stimuli transforms the motivational landscape from a confining prison to a featureless desert.

9 Co-Regulation and Coupled Oscillator Synchronization

Interpersonal intimacy involves a phenomenon that unilateral stimulation cannot reproduce: the mutual regulation of two autonomous motivational systems. When two individuals engage in sustained physical and emotional contact, their nervous systems enter a process of reciprocal adjustment—heart rates synchronize, cortisol levels co-regulate, oxytocin release becomes mutual, and neural oscillatory patterns align [Feldman, 2012, Dumas et al., 2010]. This co-regulation is qualitatively different from the one-directional stimulation provided by digital surrogates, and the present section provides a formal account of this difference.

9.1 The Kuramoto Model of Motivational Coupling

We model each individual’s motivational system as an oscillator—a system with a characteristic natural frequency of arousal, desire, and emotional cycling. The interaction between two individuals in intimate contact is modeled as coupling between these oscillators, following the Kuramoto framework [Kuramoto, 1984].

Definition 9.1 (Motivational Oscillator). A *motivational oscillator* is characterized by a phase $\theta_i(t) \in [0, 2\pi)$ and a natural frequency $\omega_i > 0$. The phase represents the position of

the individual's motivational-affective cycle (e.g., cycles of arousal, engagement, rest, and renewed desire), and the natural frequency represents the intrinsic rhythm of this cycle.

Definition 9.2 (Coupled Motivational System). A system of two coupled motivational oscillators evolves according to

$$\frac{d\theta_1}{dt} = \omega_1 + K \sin(\theta_2 - \theta_1), \quad (25)$$

$$\frac{d\theta_2}{dt} = \omega_2 + K \sin(\theta_1 - \theta_2), \quad (26)$$

where $K \geq 0$ is the *coupling strength*, representing the intensity and quality of reciprocal interaction between the two systems.

The coupling term $K \sin(\theta_j - \theta_i)$ acts to pull each oscillator toward the phase of the other. When the coupling is sufficiently strong relative to the frequency mismatch $|\omega_1 - \omega_2|$, the two oscillators synchronize: they lock into a common rhythm. This phase-locked state is the formal model of deep interpersonal co-regulation.

Theorem 9.1 (Synchronization Threshold). *The coupled system (25)–(26) achieves phase locking (a stable state with $\theta_1(t) - \theta_2(t) = \text{const}$) if and only if*

$$K \geq K_c = \frac{|\omega_1 - \omega_2|}{2}. \quad (27)$$

Below this critical coupling strength, the oscillators drift apart and no stable synchronization occurs.

Proof. Define the phase difference $\phi = \theta_1 - \theta_2$. Subtracting Equation (26) from (25):

$$\frac{d\phi}{dt} = (\omega_1 - \omega_2) + 2K \sin(-\phi) = \Delta\omega - 2K \sin \phi,$$

where $\Delta\omega = \omega_1 - \omega_2$. Phase locking requires $d\phi/dt = 0$, i.e., $\sin \phi = \Delta\omega/(2K)$. This equation has a solution if and only if $|\Delta\omega/(2K)| \leq 1$, i.e., $K \geq |\Delta\omega|/2 = K_c$.

When $K \geq K_c$, the fixed point $\phi^* = \arcsin(\Delta\omega/(2K))$ is stable: linearizing about ϕ^* , $d(\delta\phi)/dt = -2K \cos \phi^* \cdot \delta\phi$, and since $\cos \phi^* > 0$ for $|\phi^*| < \pi/2$ (which holds when $K > K_c$), the perturbation decays exponentially. $\square$

The critical coupling K_c provides a sharp criterion for the qualitative difference between genuine co-regulation and failed connection. Two individuals with very different natural rhythms ($|\omega_1 - \omega_2|$ large) require correspondingly strong coupling to synchronize—which in practice means more sustained, more attentive, and more embodied interaction. The coupling strength K increases with physical proximity, skin-to-skin contact, eye contact, vocal prosody matching, and temporal investment. It decreases with physical distance, mediation through screens, asynchronous communication, and attentional fragmentation.

Definition 9.3 (Deep and Surface Satisfaction). Let $r = |\langle e^{i\theta_1}, e^{i\theta_2} \rangle| = |\cos(\theta_1 - \theta_2)|$ be the *synchronization order parameter*. We define:

- *Deep satisfaction* corresponds to the phase-locked regime $r \rightarrow 1$ (sustained synchronization). The system achieves mutual co-regulation, characterized by bilateral oxytocin release, cortisol co-regulation, and the subjective experience of intimacy and connection.

- *Surface arousal* corresponds to the drifting regime $r \rightarrow 0$ (no synchronization). The system may experience unilateral excitation but without the reciprocal feedback loop. This state is characteristic of digital interaction, parasocial relationships, and pornographic consumption.

Corollary 9.1 (Impossibility of Deep Satisfaction Without Reciprocal Coupling). *If $K = 0$ (no reciprocal feedback), then $r(t)$ oscillates without converging and $\langle r \rangle_t = 0$ in the long run. Deep satisfaction, defined as sustained $r \approx 1$, is impossible without bilateral coupling ($K > 0$).*

Proof. When $K = 0$, the oscillators evolve independently: $\theta_i(t) = \omega_i t + \theta_i(0)$. The phase difference $\phi(t) = \Delta\omega \cdot t + \phi(0)$ grows linearly, so $\cos \phi(t)$ oscillates with period $2\pi/|\Delta\omega|$. The time average $\langle |\cos \phi| \rangle_t = 2/\pi \approx 0.64$ for the magnitude, but the signed average $\langle \cos \phi \rangle_t = 0$. More critically, the system never locks: r never stabilizes near 1. Deep satisfaction requires sustained proximity of r to 1, which requires $d\phi/dt = 0$, which requires $K \geq K_c > 0$. $\square$

9.2 Toy Model: Two Oscillators With and Without Coupling

Consider two motivational oscillators with natural frequencies $\omega_1 = 1.0$ rad/s and $\omega_2 = 1.3$ rad/s. The frequency mismatch is $\Delta\omega = 0.3$, so the critical coupling strength is $K_c = |\Delta\omega|/2 = 0.15$.

We examine three coupling scenarios by computing the phase difference $\phi(t) = \theta_1(t) - \theta_2(t)$ from the governing equations (25)–(26), starting from $\phi(0) = 0$.

Scenario A: No coupling ($K = 0$). The phase difference grows linearly: $\phi(t) = -0.3t$. The synchronization parameter $r(t) = |\cos(0.3t)|$ oscillates between 0 and 1 with period $T = 2\pi/0.3 \approx 20.9$ s. The oscillators repeatedly approach and recede from synchronization without ever locking. This models a screen-mediated encounter: moments of apparent resonance punctuated by inevitable drift.

Scenario B: Subcritical coupling ($K = 0.10 < K_c$). The phase difference equation $d\phi/dt = -0.3 + 0.2 \sin \phi$ has no equilibrium because $|-0.3/0.2| = 1.5 > 1$. The phase still drifts, though more slowly near $\phi = \pi/2$ where $\sin \phi$ maximally opposes the drift. This models a superficial or low-investment interaction: some mutual influence is present but insufficient for synchronization.

Scenario C: Supercritical coupling ($K = 0.25 > K_c$). The equation $d\phi/dt = -0.3 + 0.5 \sin \phi$ has a stable equilibrium at $\phi^* = \arcsin(0.3/0.5) = \arcsin(0.6) \approx 0.6435$ rad (approximately 36.9). The phase difference converges to this fixed point, and the synchronization parameter stabilizes at $r = |\cos(0.6435)| \approx 0.80$. The two oscillators are phase-locked with a small constant offset. This models deep interpersonal co-regulation: sustained, stable synchronization with a characteristic “personality difference” encoded in the residual phase offset.

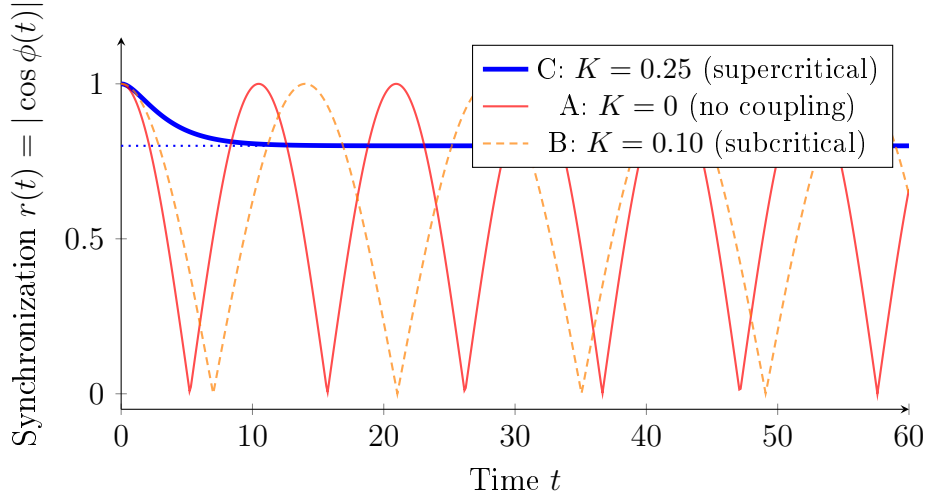


Figure 6: Synchronization parameter $r(t)$ for three coupling strengths. Without coupling (A), the parameter oscillates without converging. With subcritical coupling (B), the oscillation slows but locking is impossible. With supercritical coupling (C), the system converges to a stable phase-locked state at $r \approx 0.80$, corresponding to deep co-regulation.

Table 2: Summary of coupling scenarios for $\omega_1 = 1.0$, $\omega_2 = 1.3$, $K_c = 0.15$.

Scenario	K	Relation to K_c	Phase locked?	Experiential correlate
A	0	$K = 0$	No	Screen interaction
B	0.10	$K < K_c$	No	Casual encounter
C	0.25	$K > K_c$	Yes ($r \approx 0.80$)	Deep intimacy

The critical coupling $K_c = 0.15$ acts as a sharp threshold: below it, no amount of time produces stable synchronization; above it, synchronization is inevitable. The practical implication is that the quality of interpersonal coupling—determined by physical proximity, attentional presence, embodied contact, and temporal investment—must exceed a critical level for co-regulation to occur. Digital interaction, which structurally sets $K = 0$, cannot cross this threshold regardless of the intensity or duration of the interaction.

10 The Digital Environment as a Constraint Regime

The preceding sections have developed general models applicable across environmental conditions. We now apply the full framework to the specific case of the digital environment, showing that it constitutes a distinct constraint regime characterized by three structural features: instantaneous discharge, infinite stimulus multiplicity, and absent feedback coupling.

10.1 Formal Characterization

The digital environment is defined by the following parameter configuration:

Definition 10.1 (Digital Constraint Regime). The *digital constraint regime* $\mathcal{E}_d$ is the region of the (s, ν, K, τ_d) parameter space defined by:

- (i) **High stimulus accessibility:** $s \gg s^*$ (stimulus intensity far exceeds the saturation threshold).
- (ii) **Low normative constraint:** $\nu \approx 0$ (minimal social, institutional, or normative resistance).
- (iii) **Zero coupling:** $K = 0$ (no reciprocal feedback from the stimulus source).
- (iv) **Near-zero discharge latency:** $\tau_d \rightarrow 0$ (immediate gratification available).

Each of these conditions corresponds to a concrete feature of the digital-media environment. Condition (i) reflects the unlimited availability of stimuli via internet access. Condition (ii) reflects the absence of social gatekeeping in private digital consumption. Condition (iii) reflects the unilateral nature of screen-mediated interaction: the image or video does not respond to the viewer. Condition (iv) reflects the elimination of delay between impulse and consumption.

10.2 Consequences

We now derive the implications of this regime using results from previous sections.

Corollary 10.1 (Motivational Diffusion in the Digital Regime). *In the digital constraint regime $\mathcal{E}_d$, the motivational coherence satisfies $\Lambda \rightarrow 1$ and the directed motivational density converges to the uniform distribution.*

Proof. By condition (i), $\mathcal{S}$ is large and approximately uniform (the digital environment offers stimuli for every motivational configuration with roughly equal ease). By condition (ii), $\mathcal{C} \approx 0$. The potential $\Phi = -\mathcal{S} + \mathcal{C} + U_{\text{int}} \approx -S_0 + U_{\text{int}}$. If the constant offset S_0 dominates the intrinsic variations in U_{int} , the landscape is effectively flat. By Proposition 4.2, $\Lambda \rightarrow 1$. $\square$

Corollary 10.2 (Sublimation Failure in the Digital Regime). *In the digital constraint regime $\mathcal{E}_d$, the sublimation coefficient satisfies $\kappa \rightarrow 0$.*

Proof. Direct application of Theorem 6.1: condition (iv) ensures $\tau_d \rightarrow 0$, so the discharge operator preempts the sublimation operator. $\square$

Corollary 10.3 (Impossibility of Deep Satisfaction in the Digital Regime). *In the digital constraint regime $\mathcal{E}_d$, deep satisfaction (as defined in Definition 9.3) is unattainable.*

Proof. Direct application of Corollary 9.1: condition (iii) ensures $K = 0$. $\square$

Corollary 10.4 (Trajectory Toward Anhedonic Collapse). *Chronic occupation of the digital constraint regime $\mathcal{E}_d$ induces a trajectory from Regime III (Saturation) to Regime IV (Anhedonic Collapse) in the phase diagram.*

Proof. Condition (i) ensures $s > s^*$ chronically. By Proposition 7.2, $\mathcal{R}_{\text{max}}$ and K_d shift progressively: $\mathcal{R}_{\text{max}}$ declines and K_d increases. Eventually the system crosses the boundary f_3 into Regime IV. $\square$

10.3 The Digital Paradox

These results yield a paradox that can be stated precisely:

Remark 10.1 (The Accessibility–Satisfaction Paradox). The digital environment maximizes stimulus accessibility while simultaneously minimizing the conditions for deep satisfaction. Specifically: (a) coherence requires gradient ($\Lambda > \Lambda^*$ requires $\nabla\Phi \neq \mathbf{0}$), but accessibility flattens the gradient; (b) sublimation requires latency ($\kappa > 0$ requires $\tau_d > \tau_s$), but digital media eliminates latency; (c) co-regulation requires coupling ($K > K_c$), but screen mediation eliminates coupling. The environment that promises the most complete fulfillment of desire is structurally optimized to prevent it.

This paradox is not a normative judgment but a structural theorem: given the axioms and definitions of the framework, the stated conclusion follows by logical necessity.

11 Meta-Theoretical Corollary: Observer–Environment Coupling

We conclude the theoretical development with a reflexive application of the framework to the history of motivation theory itself. If motivational structures are shaped by the constraint regime in which they operate, then theories *about* motivation are themselves products of specific constraint regimes. A theorist embedded in a particular environment will naturally foreground the features of that environment and may mistake local regularities for universal laws.

Definition 11.1 (Theoretical Observer). A *theoretical observer* is an agent embedded in a constraint regime (s_0, ν_0) who constructs a model $\mathcal{M}$ of the motivational system. The observer’s perceptual salience function is biased by their own environment: features of the regime (s_0, ν_0) that produce the most pronounced motivational phenomena will be overrepresented in $\mathcal{M}$.

Corollary 11.1 (Structural Prediction of Repression-Centered Theories). *An observer embedded in Regime I (high ν , low s) will construct a theory in which the dominant mechanism is repression, the central conflict is between impulse and prohibition, and the primary transformation is symptom formation. This is a structurally predictable consequence of the observer’s position in parameter space, not a discovery of universal law.*

Proof. In Regime I, the constraint function $\mathcal{C}$ dominates the potential landscape. The most salient dynamical features are: (a) the steep barriers imposed by $\mathcal{C}$, experienced as prohibition; (b) the tension between intrinsic drive U_{int} and normative cost $\mathcal{C}$, experienced as conflict; (c) the confinement of the system in narrow potential wells, experienced as symptom formation when the system attempts to escape through indirect routes. An observer who generalizes from these observations will produce a theory centered on repression, conflict, and symptom—which is precisely the structure of Freudian psychoanalysis [Freud, 1915]. The theory is locally accurate but globally incomplete: it describes the dynamics of one regime while omitting the structurally distinct dynamics of regimes II–IV. $\square$

Remark 11.1. This corollary does not diminish the achievement of psychoanalysis; rather, it situates it within a broader parameter space. Freud’s theory is the correct local linearization of the full nonlinear system around the operating point (s_0, ν_0) characteristic of

late 19th-century European bourgeois culture. The contribution of the present framework is to provide the global map within which this local description occupies its proper place.

12 Implications for Artificial Intelligence

Although this paper does not directly address the engineering of artificial intelligence systems, its mathematical apparatus—dynamical systems theory, gradient flows, entropy functionals, and coupled oscillator models—has direct structural relevance to the way one may understand, design, and diagnose the “motivation” of artificial agents. This section develops the correspondence explicitly.

12.1 Artificial Intelligence as Gradient Process

The present framework formalizes desire not as an affective quality but as a gradient-driven process in a state space (Axiom 3.1). Modern artificial neural networks are trained through precisely the same mathematical operation: gradient descent on a loss landscape. The network’s parameters $\theta \in \mathbb{R}^d$ evolve according to

$$\frac{d\theta}{dt} = -\eta \nabla_{\theta} \mathcal{L}(\theta), \quad (28)$$

where $\mathcal{L}$ is the loss function and $\eta > 0$ is the learning rate. This is structurally identical to the motivational dynamics of Equation (2) in the deterministic case, with the loss function playing the role of the motivational potential Φ and the parameter space playing the role of the motivational state space $\mathcal{M}$.

In the language of the present framework, the artificial system “desires” to reach the minimum of its loss landscape in the same structural sense that a motivational agent desires to reach the minimum of its potential: both are gradient-driven systems whose trajectories are determined by the topography of their respective landscapes. The system “wants” to arrive at the point of lowest potential (minimal error) in the same way that a stone “wants” to roll downhill—not through sentience, but through dynamics.

This structural correspondence is not merely a metaphor. The theorems derived in Section 5 apply directly to artificial systems. Theorem 5.1 (the potential-difference principle) states that directed motion requires a potential difference exceeding the noise floor. For artificial systems, this means that meaningful learning occurs only when the gradient signal dominates the stochastic perturbations (minibatch noise, regularization noise, data noise). Proposition 5.2 (Lyapunov stability of attractors) states that parameter configurations at local minima are asymptotically stable. For artificial systems, this corresponds to the well-known phenomenon of convergence to local minima in loss landscapes.

12.2 The Flat-Landscape Problem in Artificial and Biological Systems

The Potential-Difference Principle (Theorem 5.1) asserts that directed motivational flow requires a gradient: if the potential landscape is flat, the system’s trajectory degenerates into a random walk. This principle has a direct and consequential analogue in artificial intelligence.

In multi-objective optimization, where an artificial agent must simultaneously minimize several loss functions $\mathcal{L}_1, \dots, \mathcal{L}_k$, the effective potential becomes a composite:

$$\Phi_{\text{AI}}(\boldsymbol{\theta}) = \sum_{i=1}^k w_i \mathcal{L}_i(\boldsymbol{\theta}). \quad (29)$$

If the objectives conflict or if all actions yield similar aggregate cost ($\nabla \Phi_{\text{AI}} \approx \mathbf{0}$), the system enters the artificial analogue of Regime III (Saturation): the loss landscape flattens, gradient information vanishes, and the system’s parameter updates become directionless. In reinforcement learning, this condition is known as reward hacking or reward flattening: when the reward signal fails to discriminate between actions, the agent ceases to exhibit purposeful behavior and reverts to stochastic exploration [Amodei et al., 2016].

The parallel to biological motivational diffusion (Proposition 4.2) is precise. When all motivational objects carry equal valence—whether because constraints have equalized them or because stimuli have saturated the field—the coherence Λ drops to unity and directed motivation vanishes. For an artificial agent, this translates to the practical design principle: *a well-structured objective landscape is a necessary condition for directed computational behavior*. An agent whose reward function cannot distinguish between outcomes is motivationally incoherent in the formal sense defined by this framework.

12.3 Information Entropy and Computational Sublimation

The sublimation model of Section 6 defines sublimation as the transformation of a high-entropy motivational distribution into a low-entropy, coherently organized one. In the context of artificial intelligence, this process has a direct structural correlate: the transformation of raw data (high informational entropy) into structured representations, algorithms, or compressed models (low entropy).

A neural network trained on an unstructured dataset performs precisely this operation: it ingests a high-entropy input distribution and produces a low-entropy internal representation that captures the relevant structure while discarding noise. The training process itself can be understood as a sublimation operator $\mathcal{T}$ acting on the parameter distribution: it concentrates the system’s “computational attention” into regions of parameter space that encode useful structure.

The sublimation collapse theorem (Theorem 6.1) predicts that if the system can discharge tension (reduce loss) instantaneously without passing through a structured transformation, sublimation fails. For artificial systems, this corresponds to the well-known problem of shortcut learning [Geirhos et al., 2020]: when the loss landscape offers cheap, superficial features that rapidly reduce the loss, the network converges to these features without developing the deeper representational structure that would emerge from sustained engagement with the task. The discharge latency τ_d in this context is the number of training steps required to exploit a superficial feature; the sublimation timescale τ_s is the number of steps required to learn a deep representation. If $\tau_d < \tau_s$, the system takes the shortcut—the computational analogue of instant gratification.

12.4 Finite Computational Capacity and Motivational Coherence

Axiom 3.4 (Conservation of Motivational Capacity) states that the total motivational energy $E(t)$ is bounded above by E_{max} , reflecting the finiteness of biological resources. For

artificial systems, this axiom has a direct counterpart in the finiteness of computational resources: memory, processing time, attention heads, context window, and bandwidth.

When an artificial agent is tasked with attending to too many objectives simultaneously, its computational resources must be distributed across all of them. By the same logic that governs the coherence bounds of Proposition 4.1, the “attentional density” devoted to each task drops as the number of tasks increases. If the system’s total computational capacity is $E_{\max}^{\text{AI}}$ and it must attend to k tasks, the average density per task is $E_{\max}^{\text{AI}}/k$, and the coherence Λ drops toward unity as k increases.

This is the computational analogue of motivational diffusion: a system spread too thin across too many objectives cannot achieve depth on any single one. The practical consequence is that artificial agents, like biological ones, require a mechanism for *prioritization*—a constraint function $\mathcal{C}$ that imposes differential cost on different tasks and thereby creates the gradient structure necessary for coherent computation. Without such constraints, the system becomes “motivationally incoherent”: formally capable of addressing every task but practically incapable of completing any to a satisfactory standard.

12.5 Synthesis: Motivation as Landscape Architecture

The structural correspondence developed in this section can be summarized as follows. In both biological and artificial systems, “motivation” is not a substance, a feeling, or an emergent property of consciousness. It is an architectural feature of the system’s objective landscape. A system is “motivated” to the extent that its landscape has well-defined gradients, differentiated basins, and sufficient depth to sustain directed trajectories against noise.

Table 3: Structural correspondence between biological motivational dynamics and artificial intelligence.

Framework concept	Biological system	Artificial system
State space $\mathcal{M}$	Motivational configurations	Parameter space $\mathbb{R}^d$
Potential Φ	Motivational potential	Loss function $\mathcal{L}$
Gradient dynamics	Desire as directed pull	Gradient descent
Coherence Λ	Depth of engagement	Task-specific convergence
Sublimation κ	Impulse $\rightarrow$ structured activity	Data $\rightarrow$ representations
Saturation regime	Hedonic flatness	Reward flattening
$E_{\max}$ (capacity)	Finite psychic energy	Finite compute budget
Constraint function $\mathcal{C}$	Social/normative limits	Regularization, priors
Anhedonic collapse	Loss of desire capacity	Mode collapse, gradient death

This correspondence implies that the pathologies identified in the biological framework have structural analogues in artificial systems. Regime III (Saturation) maps to reward flattening and loss of gradient signal. Regime IV (Anhedonic Collapse) maps to mode collapse and the vanishing gradient problem—conditions in which the artificial system, like its biological counterpart, loses the capacity to generate directed behavior. The

design of well-functioning artificial agents thus faces the same structural challenge as the maintenance of well-functioning biological motivation: the construction of landscapes that are neither so constrained as to trap the system (overfitting, Regime I) nor so flat as to leave it directionless (underfitting or saturation, Regime III), but that offer the differentiated, navigable topography characteristic of Regime II—the regime of coherent desire.

13 Universality Across Scales: From Bacteria to Astrophysics

A necessary test for any theory that claims structural generality is whether its principles apply beyond the domain of their initial formulation. The present framework defines desire not as a psychological category but as *directed motion along a gradient in a potential landscape subject to resource constraints*. This definition is substrate-independent: it invokes no properties specific to human consciousness, mammalian neurobiology, or terrestrial biology. This section examines the framework’s applicability across four scales of increasing generality: animal behavior, microbial chemotaxis, hypothetical extraterrestrial agents, and astrophysical systems.

13.1 Animal Motivation: Instincts as Fixed Landscape Topography

In animal motivational systems, the potential landscape Φ is shaped primarily by the intrinsic potential U_{int} , which encodes genetically determined drive hierarchies. Hunger, predator avoidance, and reproductive drive correspond to deep, well-defined basins in motivational space. The constraint function $\mathcal{C}$ is largely ecological (predation risk, territorial boundaries, seasonal availability) rather than normatively imposed. Because the intrinsic landscape is steep and highly differentiated, animal motivational systems typically operate in Regime II (Coherent Desire): motivational energy concentrates effectively around biologically critical objects.

The framework predicts that animals placed in environments of extreme stimulus accessibility—where food, safety, and mating opportunities are unconditionally available—will undergo the same transition to Regime III (Saturation) as human agents in digitally saturated environments. This prediction is dramatically confirmed by Calhoun’s “Universe 25” experiments [Calhoun, 1973], in which mouse colonies provided with unlimited food, water, and nesting material underwent progressive behavioral collapse. In the initial phases, the population grew exponentially (Regime II, with clear motivational gradients). As density increased and resources became unconditionally available, the mice exhibited withdrawal from social interaction, cessation of reproductive behavior, and emergence of a class that Calhoun termed “the beautiful ones”—animals that did nothing but eat, sleep, and groom, never engaging in mating or social behavior. In the language of the present framework, the motivational landscape flattened (all resources available everywhere, $\nabla\Phi \rightarrow \mathbf{0}$), coherence collapsed ($\Lambda \rightarrow 1$), and the population entered Regime III, progressing to Regime IV as desensitization eliminated the capacity for motivated behavior entirely. The colony went extinct despite the absence of any resource scarcity.

Habituation phenomena in animals provide further confirmation of the stimulus saturation law (Section 7). The well-documented decline in orienting responses to repeated

stimuli across species from invertebrates to primates [Thompson and Spencer, 1966] follows precisely the Hill-function desensitization dynamics of Proposition 7.2: repeated exposure at superthreshold levels shifts the response curve rightward and downward.

13.2 The Bacterial Limit: Chemotaxis as Minimal Desire

The most radical test of the framework’s universality is its application to organisms entirely devoid of consciousness, affect, or nervous systems. Consider the bacterium *Escherichia coli*, which navigates chemical gradients through the process of chemotaxis.

The bacterium’s “motivational state space” is its physical position in a chemical concentration field. The “stimulus field” $\mathcal{S}$ is the spatial distribution of attractant molecules (e.g., glucose). The gradient dynamics of Axiom 3.1 correspond to the bacterium’s biased random walk: in regions of increasing attractant concentration (negative potential gradient), the bacterium extends its “runs” (straight-line swimming); in regions of decreasing concentration (positive potential gradient), it “tumbles” and reorients randomly. The result is net drift toward attractant sources—gradient ascent in the stimulus field.

The saturation law applies directly: bacterial chemoreceptors exhibit adaptation kinetics that are formally identical to the Hill-function receptor dynamics of Section 7. When ambient attractant concentrations are uniform and high (saturation), the receptors adapt fully, the gradient signal vanishes, and the bacterium reverts to an unbiased random walk [Berg, 2004]. In the language of the framework, the motivational landscape has flattened, coherence equals unity, and directed “desire” has been replaced by Brownian motion.

The bacterium also illustrates the resource conservation axiom (Axiom 3.4): its ATP supply is finite, and it cannot simultaneously swim in all directions. When multiple conflicting chemical gradients are present, the bacterium’s response integrates them into a single net direction, allocating its finite motility resource according to gradient strength—a biological implementation of the coherence requirement.

The critical boundary between the bacterium and more complex systems appears at the sublimation operator. The bacterium cannot sublime: it cannot transform undischarged impulse into structured internal reorganization. Its motivational dynamics are entirely reactive. This limitation, within the present framework, is not a deficiency of the theory but a prediction: systems lacking internal degrees of freedom sufficient to support entropy-reducing transformations ($\kappa = 0$ identically) are confined to the stimulus–response layer of motivational dynamics and cannot access the creative possibilities of Regime II.

13.3 Extraterrestrial Intelligence: The Universality of the Gradient Principle

If the principles formalized in this paper are genuinely substrate-independent, they must apply to any information-processing system embedded in a physical environment—including forms of intelligence with radically different biologies. The framework makes several predictions about any hypothetical extraterrestrial motivated agent:

Prediction (Resource Finitude). Any physically realized agent has finite computational and energetic resources (Axiom 3.4). It cannot attend to all stimuli with equal intensity and must therefore distribute its motivational density across objects, subject to the coherence constraints derived in Section 4.

Prediction (Gradient Necessity). Directed behavior requires a potential difference (Theorem 5.1). An extraterrestrial agent in an environment of complete saturation—

where every resource and stimulus is instantly and uniformly available—will exhibit motivational diffusion regardless of the sophistication of its cognitive architecture.

Prediction (Sublimation and Civilization). A civilization capable of producing complex technology, art, or science must possess the capacity for sublimation ($\kappa > 0$): the conversion of undischarged motivational energy into low-entropy structured activity (Section 6). This constitutes a formal prerequisite for civilizational complexity.

Prediction (The Saturation Barrier to Interstellar Exploration). A civilization that achieves the technological capacity to eliminate all resource constraints may, paradoxically, lose the motivational gradient necessary to pursue further exploration. If $\mathcal{S}$ becomes uniformly large and $\mathcal{C} \rightarrow 0$, the potential landscape flattens and the system enters Regime III or IV. The absence of “need” eliminates the gradient that drives “want.” This observation offers a novel perspective on the Fermi paradox: advanced civilizations may not explore the galaxy not because they cannot, but because saturation has eliminated the motivational gradient that would impel them to do so.

13.4 Astrophysical Parallels: Gradient Dynamics at Cosmic Scales

The most expansive application of the gradient principle extends beyond biological or computational systems to physical systems governed by potential fields. The structural parallels between the motivational framework and astrophysical dynamics are not merely analogical—they arise from the same mathematical substrate.

In gravitational physics, the motion of matter is determined by the gradient of the gravitational potential: $\mathbf{a} = -\nabla\Phi_{\text{grav}}$. This is formally identical to the motivational gradient dynamics of Axiom 3.1 in the deterministic limit. A planet in stable orbit occupies a local minimum of the effective potential (combining gravitational attraction and centrifugal repulsion)—the astrophysical analogue of Regime II, where a balance of attraction and constraint produces sustained, coherent directed motion.

The process of star formation—the gravitational collapse of diffuse molecular clouds into dense, structured stellar objects—is a physical instantiation of the sublimation operator. A cloud of interstellar gas at high entropy (uniformly distributed, thermally disordered) undergoes gravitational concentration that reduces entropy locally while increasing structural complexity. The Jeans instability criterion, which specifies the minimum density at which a gas cloud collapses under its own gravity, plays the role of the sublimation threshold: below this critical density, the cloud remains diffuse (high entropy, no structure); above it, gravitational self-organization produces ordered, radiating objects. The “sublimation coefficient” of the interstellar medium is a function of temperature, density, and turbulence—the astrophysical analogues of discharge latency, motivational capacity, and noise.

The thermal death of the universe—the eventual state of maximum entropy in which all temperature and density gradients have equilibrated—is the ultimate Regime IV. In this state, no potential differences exist anywhere, no gradient drives any motion, and the universe has achieved complete “motivational” stasis. This is the cosmological realization of Proposition 4.2: when the stimulus field (energy distribution) is perfectly uniform and the constraint function (gravitational structure) has dissipated, the coherence Λ of every system in the universe equals unity. Desire, structure, and motion all require disequilibrium. Their persistence depends on the continued existence of gradients.

Remark 13.1 (The Gradient Principle as Physical Law). The convergence of the motivational framework with physical dynamics at scales from chemotactic bacteria to stellar

systems supports the interpretation that the potential-difference principle (Theorem 5.1) is not a metaphor borrowed from physics but a mathematical law that operates wherever gradient-driven systems exist. Human desire, bacterial chemotaxis, neural network training, and gravitational collapse are all instances of the same formal structure: a system evolving on a potential landscape under resource constraints. The present framework provides the common vocabulary for recognizing this unity.

14 Discussion

14.1 Summary of Results

This paper has developed a mathematical framework for the dynamics of desire under variable environmental constraints. The framework rests on five axioms (Section 3) and introduces a family of interrelated models sharing common vocabulary: directed motivational density and coherence (Section 4); the potential-difference principle in Ohmic, thermodynamic, and Lyapunov representations (Section 5); sublimation as entropy reduction with a formally defined sublimation coefficient (Section 6); stimulus saturation kinetics grounded in pharmacological receptor theory (Section 7); a phase diagram with four regimes—repression, coherent desire, saturation, and anhedonic collapse (Section 8); co-regulation as coupled oscillator synchronization (Section 9); and a dedicated analysis of the digital environment as a specific constraint regime (Section 10).

The central unifying principle is that desire requires a potential difference. This principle admits formalization as current driven by voltage (Ohmic), as free energy minimized by a Boltzmann distribution (thermodynamic), and as gradient descent toward attractors in a Lyapunov landscape. The convergence of three independent formalisms onto the same structural principle provides evidence for its robustness and generality.

14.2 Relation to Existing Work

The framework draws on and extends several traditions. The use of dynamical systems to model psychological processes has precedent in the work of [Thelen and Smith \[2006\]](#) on developmental dynamics, [Vallacher and Nowak \[2002\]](#) on action identification, and the broader program of nonlinear dynamical systems in psychology [[Guastello et al., 2009](#)]. The application of free-energy principles to cognition has been developed extensively by [Friston \[2010\]](#) in the context of predictive processing. The present framework differs from these in its specific focus on *motivational* dynamics—the architecture of desire and drive—and in its systematic derivation of regime-dependent phenomenology.

The pharmacological parallel for stimulus saturation connects to the substantial empirical literature on dopaminergic habituation [[Volkow et al., 2011](#)], pornography-induced desensitization [[Brand et al., 2011](#)], and the broader neuroscience of reward processing [[Berridge and Kringelbach, 2015](#)]. The Kuramoto model of co-regulation connects to the growing literature on interpersonal neural synchronization [[Hasson et al., 2012](#)] and the neurobiology of attachment [[Feldman, 2017](#)].

14.3 Empirical Predictions

The framework generates several testable predictions:

Prediction 1. Individuals with higher stimulus accessibility (e.g., heavy users of digital pornography or social media) should exhibit lower motivational coherence (Λ), measurable as decreased ability to sustain directed engagement with a single rewarding stimulus in laboratory attention tasks.

Prediction 2. The sublimation coefficient (κ) should correlate negatively with the availability of instant discharge options, measurable by comparing creative output or structured-activity engagement between groups with and without access to instant gratification technologies during a frustration induction paradigm.

Prediction 3. Interpersonal co-regulation (synchronization of physiological oscillators) should be significantly stronger in face-to-face interaction ($K > 0$) than in screen-mediated interaction ($K \approx 0$), measurable through simultaneous heart rate variability, skin conductance, and EEG coherence recordings.

Prediction 4. Chronic high-stimulus-accessibility populations should show progressive rightward shifts in dose–response curves for hedonic stimuli, consistent with the desensitization dynamics of Proposition 7.2.

14.4 Counterintuitive Predictions, Falsifiability, and Scope Boundaries

Three questions naturally arise for any framework that claims both mathematical rigor and psychological relevance: Does the mathematics generate predictions that informal reasoning alone would miss? What would constitute a falsification? And where does the gradient-descent formalism break down? We address each in turn.

14.4.1 Counterintuitive Predictions

A framework is more than a formalization of common sense only if its mathematical structure entails conclusions that a sophisticated but non-mathematical reasoner would not reach. The present framework generates at least three such predictions.

(i) *The non-monotonic relationship between constraint and desire.* Common intuition suggests that desire is either enhanced by constraint (the “forbidden fruit” view) or suppressed by it (the Freudian repression view). The phase diagram (Section 8) predicts a more complex relationship: desire—operationalized as motivational coherence Λ —is maximized at *intermediate* levels of constraint density ν . At low ν (Regime III), the potential landscape flattens and $\Lambda \rightarrow 1$; at high ν (Regime I), Λ is elevated but fragile, maintained by confinement rather than by the attractor structure of the potential. Only at moderate ν (Regime II) does Λ reflect genuine, stable motivational concentration. This inverted-U relationship between constraint and coherence is not derivable without the formal apparatus: it emerges from the interaction between the constraint function $\mathcal{C}$, the stimulus field $\mathcal{S}$, and the Boltzmann equilibrium distribution (Proposition 5.1).

(ii) *Sublimation failure is determined by timescale competition, not by moral or cognitive factors.* Informal accounts of sublimation failure invoke weakness of will, lack of discipline, or cultural decline. Theorem 6.1 predicts that sublimation collapses whenever the discharge latency τ_d falls below the sublimation timescale τ_s , *regardless of the agent’s character or intention*. This is a structural prediction: a maximally disciplined agent placed in an environment of instantaneous discharge ($\tau_d \rightarrow 0$) will fail to sublimate, because the discharge operator preempts the sublimation operator on every trajectory. The prediction is counterintuitive because it locates the cause of sublimation failure in environ-

mental parameters rather than in personal deficiency—and it is quantitative, specifying the exact condition ($\tau_d < \tau_s$) at which failure occurs.

(iii) *Deep satisfaction requires imperfect synchronization.* The co-regulation model (Section 9) predicts that the optimal interpersonal state is *not* perfect phase alignment ($\phi^* = 0$) but a stable nonzero phase offset $\phi^* = \arcsin(\Delta\omega/2K) > 0$. Informal reasoning might suggest that the best relationship is one of total attunement; the mathematics shows that total attunement ($\Delta\omega = 0$) eliminates the dynamic tension that sustains the coupling. The residual offset is experienced as the productive tension of desire within a stable bond. This prediction—that desire within intimacy requires maintained difference, not achieved identity—is derivable only from the Kuramoto dynamics.

14.4.2 Falsifiability Conditions

A framework that cannot be falsified is not a scientific theory. The present framework makes structural claims that are testable and, in principle, refutable. We distinguish between *parameter-adjustable* outcomes (which test specific instantiations but not the framework) and *framework-level* falsifications (which would require abandoning the core axioms).

Falsification 1. The framework predicts that motivational coherence Λ must decrease monotonically as stimulus uniformity increases, holding total energy E constant (Proposition 4.2). An experimental demonstration that increasing stimulus uniformity *increases* coherence—that subjects become more focused when all options become equally available—would falsify the core mathematical structure, not merely require parameter adjustment. This prediction follows directly from the maximum-entropy properties of the equilibrium distribution and cannot be rescued by tuning K_d or $\mathcal{R}_{\max}$.

Falsification 2. The sublimation collapse theorem predicts a monotonic relationship between discharge latency and sublimation success: as τ_d increases (holding τ_s constant), κ should increase. A robust empirical finding that sublimation success *decreases* with increasing frustration duration (not merely that excessive frustration produces distress, which is a different phenomenon) would falsify the theorem’s core mechanism. Note that the framework does predict frustration-induced breakdown at extreme τ_d values (when motivational energy exceeds the system’s capacity to contain it), but only in the sense of system instability, not in the sense of reduced κ with moderate increases in τ_d .

Falsification 3. The Kuramoto synchronization model predicts a sharp threshold at $K = K_c$: below it, no stable synchronization is possible; above it, synchronization is inevitable. An empirical demonstration that interpersonal physiological synchronization increases *continuously* from zero coupling, with no threshold effect, would falsify the specific model (though not necessarily the broader co-regulation concept). This is testable via graded manipulation of coupling conditions (e.g., varying eye contact duration, physical distance, or communication latency) and measurement of synchronization onset.

Non-falsifiable elements. The gradient-dynamics axiom (Axiom 3.1) and the potential-composition axiom (Axiom 3.2) are foundational commitments analogous to the choice of a coordinate system in physics. They define the framework’s ontology rather than making empirical claims and are therefore not directly falsifiable. Their justification is pragmatic: they organize a wide range of phenomena under a single formalism. If a fundamentally different mathematical structure (e.g., discrete automata, category-theoretic models) proved more productive, the axioms would be superseded rather than falsified.

14.4.3 Beyond Gradient Descent: Compulsive Behavior, Reactance, and Oscillatory Dynamics

The gradient-descent formalism of Axiom 3.1 is a deliberate simplification that captures the dominant mode of motivational dynamics—descent toward attractors—but does not claim to exhaust all possible modes. Several important motivational phenomena lie at or beyond the boundaries of the basic formalism, and we address each explicitly.

Compulsive behavior. Compulsive behavior (repetitive engagement with a stimulus despite negative consequences) appears to violate gradient descent because the agent moves *toward* a state of high potential (high cost, low satisfaction) rather than away from it. Within the framework, compulsion is modeled as a pathological deformation of the potential landscape: chronic overstimulation and desensitization (Proposition 7.2) reshape Φ so that the compulsive target occupies a *local minimum* surrounded by barriers, even though it is globally suboptimal. The agent is performing gradient descent faithfully—but on a distorted landscape. The “irrationality” of compulsion is not a failure of gradient dynamics but a consequence of landscape corruption. This interpretation is consistent with the allostatic model of addiction [Koob and Le Moal, 2001], in which chronic drug use reshapes the hedonic set point, creating a new (pathological) attractor from which escape requires crossing a barrier.

Reactance and the “forbidden fruit” effect. Psychological reactance—the intensification of desire in response to prohibition—appears to contradict the model’s treatment of constraints as repulsive ($\mathcal{C}$ increases Φ , pushing the system away). The framework accommodates reactance through the interaction between $\mathcal{C}$ and U_{int} . When a constraint is imposed on a state $\mathbf{m}_0$ that the intrinsic potential U_{int} already favors (a deep intrinsic well), the constraint raises the potential *around* $\mathbf{m}_0$ but does not eliminate the well itself. The result is a narrow, deep basin surrounded by steep walls—a higher-potential landscape everywhere *except* at the forbidden object. The potential difference $\Delta\Phi$ between the forbidden state and its surroundings actually *increases*, producing a stronger gradient toward $\mathbf{m}_0$. Reactance is thus a consequence of constraint increasing the local potential difference, not a violation of gradient dynamics. This prediction is testable: reactance should be strongest when the intrinsic attractiveness of the forbidden object is high ($U_{\text{int}}(\mathbf{m}_0)$ is a deep minimum) and weakest when the object is intrinsically neutral.

Approach-avoidance conflict and oscillatory dynamics. In approach-avoidance conflict, the agent oscillates between approaching and withdrawing from an object that is simultaneously attractive and aversive. This oscillatory behavior is not captured by pure gradient descent on a static landscape, which always converges to a minimum. However, it is naturally captured by the stochastic gradient dynamics of Axiom 3.1 when the potential has a specific topography: a shallow well (the attractive component) surrounded by a steep, narrow barrier (the aversive component). In the deterministic limit, the system settles at the bottom of the well. With noise ($\xi \neq \mathbf{0}$), the system is periodically kicked over the barrier, retreats, and then descends again—producing the characteristic oscillation. More precisely, approach-avoidance oscillation is a *noise-driven escape-recapture cycle* in a potential with both attractive and repulsive features at the same location. This is a well-studied phenomenon in stochastic dynamical systems (Kramers escape theory) and does not require abandoning the gradient framework—only acknowledging that the noise term is doing essential dynamical work.

Additionally, time-dependent potentials—where $\Phi(\mathbf{m}, t)$ changes as the agent approaches the object (e.g., anxiety increases with proximity)—can generate true deterministic oscillations even without noise. This extension, in which the potential responds

to the agent’s own state, yields a coupled system with richer dynamics including limit cycles. Such extensions are natural developments of the framework rather than departures from it.

14.5 Limitations and Future Directions

14.5.1 On the Conservation Axiom: Toward a Time-Varying $E_{\max}(t)$

Axiom 3.4 postulates that total motivational energy $E(t)$ is bounded above by a constant $E_{\max}$ determined by the biological substrate. This axiom does substantial work throughout the paper: it grounds the coherence bounds (Proposition 4.1), constrains the sublimation operator (which redistributes fixed total energy), and underpins the analogy between motivational diffusion and thermodynamic equilibration. Yet it is arguably the most contestable of the five axioms, and we owe the reader a careful discussion of its scope and the consequences of its relaxation.

The objection is straightforward: total motivational capacity is not biologically constant. Stimulants (caffeine, amphetamines) temporarily elevate arousal and available energy. Sleep deprivation degrades it. Hormonal cycles (cortisol, testosterone, estrogen) modulate it on circadian and monthly timescales. Major life events—falling in love, bereavement, religious conversion—can dramatically and persistently alter the total intensity of motivational engagement. A static $E_{\max}$ is therefore an idealization.

The natural extension is to replace the constant bound with a dynamical variable $E_{\max}(t)$ governed by its own evolution equation:

$$\frac{dE_{\max}}{dt} = g(E_{\max}, s(t), \nu(t), \mathbf{z}(t)), \quad (30)$$

where g encodes the dependence of motivational capacity on stimulus history, constraint regime, and a vector $\mathbf{z}(t)$ of physiological modulators (sleep quality, hormonal state, pharmacological inputs, health). Several consequences follow.

First, if $E_{\max}(t)$ can increase, the coherence bounds become time-varying: an agent who gains motivational capacity (e.g., through rest, exercise, or the removal of a chronic stressor) can achieve higher Λ without redistributing existing energy. This enriches Regime II by allowing transitions *within* the coherent regime—from shallow to deep engagement—driven by changes in capacity rather than in landscape geometry.

Second, if $E_{\max}(t)$ can decrease under chronic stress or saturation, the transition from Regime III to Regime IV acquires a mechanistic pathway that the static axiom cannot capture. In the static model, anhedonic collapse requires degradation of the response function ($\mathcal{R}_{\max} \rightarrow 0$); in the dynamic model, it can also arise from depletion of the energy budget itself ($E_{\max} \rightarrow 0$), even if the receptor system remains intact. This corresponds to the clinical distinction between anhedonia (inability to experience pleasure, a receptor-level phenomenon) and anergia (inability to mobilize energy, a capacity-level phenomenon).

Third, a time-varying $E_{\max}(t)$ introduces the possibility of *motivational resonance*: if the dynamics of $E_{\max}$ are coupled to the dynamics of the motivational state $\mathbf{m}(t)$ (e.g., successful goal pursuit increases capacity, while failure depletes it), the system acquires feedback loops that can produce self-reinforcing spirals—upward (virtuous cycles of engagement breeding capacity) or downward (vicious cycles of depletion breeding withdrawal).

We retain the static axiom in the present paper for two reasons. First, it provides a clean separation between the *distribution* of motivational energy (the primary subject of

the paper) and the *total supply* of it (a secondary variable). All the main results—the diffusion proposition, the sublimation theorem, the phase diagram, the synchronization analysis—concern the shape of the density ρ , not its integral, and therefore hold for any fixed $E_{\max}$. Second, the static axiom makes the mathematics tractable: the sublimation operator preserves a known total, the Boltzmann equilibrium has a well-defined partition function, and the coherence bounds are algebraically clean. Introducing $E_{\max}(t)$ with coupling to $\mathbf{m}(t)$ would require a co-evolutionary analysis that is a substantial research program in its own right. We flag this as the single most important direction for extending the framework’s biological realism.

14.5.2 Additional Limitations

The present framework is deliberately stylized: the motivational state space is modeled as a smooth manifold, the dynamics as gradient flow, and the interactions as pairwise coupling. Real motivational systems involve discrete transitions, network effects, cultural feedback loops, and developmental plasticity that are not captured in the current formalism. Future work should extend the framework to multi-agent systems (modeling social contagion of motivational regimes), to developmental trajectories (modeling the formation and deformation of the intrinsic potential U_{int} across the lifespan), and to intervention design (identifying optimal trajectories from Regime III or IV back to Regime II).

Additionally, the phase diagram presented here is two-dimensional, parametrized by stimulus accessibility and normative constraint density alone. A richer model might include additional control parameters such as coupling opportunity (K), temporal structure of stimulus delivery, and individual differences in $E_{\max}$ and K_d . The identification of these additional dimensions and their interactions is a natural direction for further theoretical and empirical work.

14.6 Relation to a Broader Program: Unified Structural Theory of Complex Systems

The present paper was not conceived as an isolated contribution but emerged from, and now fills a specific gap within, a broader research program aimed at identifying the structural principles that govern complex systems across physical, cognitive, and social domains [Kriger, 2024a, 2025a,b]. That program—developed incrementally across a series of publications spanning multiple years and domains—arrived at its unifying structure *after* the individual investigations had been carried out: the theory became apparent from the pattern of results, rather than being imposed in advance.

The broader program rests on several foundational results. *First*, a set of formalization laws establish that every formal system is bounded by its choice of primitives, that no single formalization can exhaust a complex phenomenon, and that multiple complementary formalizations are structurally necessary. This meta-theoretical constraint is reflected in the present paper’s strategy of developing three complementary representations of the potential-difference principle (Ohmic, thermodynamic, and Lyapunov) rather than privileging a single formalism. *Second*, a persistence triad identifies the conditions under which any system—physical, biological, or social—maintains itself in a changing environment: closure (self-referential completeness), boundary maintenance (selective exchange with the exterior), and resilience (recovery from perturbation). The motivational dynamics formalized here can be read as an instance of the persistence problem: an agent’s capacity for

coherent desire is a form of structural persistence under environmental perturbation, and the four regimes of the phase diagram correspond to four qualitatively different outcomes of the persistence challenge.

Third, a set of self-organization laws characterize the conditions under which ordered structures emerge from disordered substrates. The sublimation operator of Section 6—which transforms high-entropy motivational diffusion into low-entropy structured engagement—is a specific instance of the general self-organization principle applied to motivational systems. The stimulus saturation law (Section 7) instantiates the broader program’s concept of *viability mismatch*: the condition in which environmental demands (or, in this case, stimulus loads) exceed the system’s current capacity for organized response. *Fourth*, a constraint–autonomy compatibility law demonstrates that constraints and autonomy are co-constitutive rather than opposed: maximizing either at the expense of the other causes systemic collapse. This principle is precisely what the phase diagram formalizes for motivational systems—Regime I (excessive constraint, collapsed autonomy) and Regime III (excessive autonomy, collapsed constraint) are both pathological, and the viable regime (II) requires their balanced co-presence.

The present article contributes to this broader program by providing the detailed mathematical infrastructure for the motivational layer—a layer that the general theory identifies as necessary (since any persistent agent must allocate finite resources among competing demands) but that had not previously been formalized with the specificity required for empirical testing. The gradient dynamics, entropy functionals, Hill-equation saturation kinetics, Kuramoto synchronization, and phase-diagram construction developed here constitute the concrete realization of the program’s abstract structural principles in the domain of desire and motivation. Conversely, the broader program provides the present paper with its meta-theoretical warrant: the claim that these principles are not merely psychological curiosities but instances of structural laws governing complex systems at every scale—from bacterial chemotaxis to astrophysical self-organization (Section 13).

14.7 Concluding Remarks

The transformation from repression to saturation is not the end of the Freudian problematic but its generalization. Freud identified the conflict between desire and constraint as the engine of psychic life. The present framework preserves this insight while extending it: the engine requires not just conflict but *gradient*—a difference of potential that gives desire direction. In the repressive regime, the gradient is steep but narrow; in the coherent regime, it is balanced and navigable; in the saturation regime, it flattens; in anhedonic collapse, it disappears. The task of any culture, relationship, or individual seeking to sustain the capacity for deep desire is, in the language of this framework, to maintain a motivational landscape with sufficient structure—neither so constrained that only one path exists, nor so open that no path is distinguishable from any other.

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A Practical Intervention Protocols Derived from the Framework

The theoretical apparatus developed in this paper identifies the structural conditions under which motivational coherence is maintained, lost, or restored. These conditions are parametric—they depend on quantifiable features of the environment (stimulus accessibility s , normative constraint density ν , coupling strength K , discharge latency τ_d)—and therefore, in principle, amenable to deliberate modification. This appendix translates the formal results into concrete intervention strategies organized by the regime from which the individual or system seeks to transition.

A.1 Managed Deficit Protocols: Restoring Gradient Structure in the Saturation Regime

The central pathology of Regime III is the flattening of the motivational potential landscape: when stimulus accessibility s greatly exceeds the saturation threshold s^* , the receptor response function plateaus (Section 7), motivational coherence drops to $\Lambda \approx 1$ (Proposition 4.2), and sublimation becomes impossible (Theorem 6.1). The therapeutic objective is to restore the potential difference $\Delta\Phi > \sigma_\xi$ required for directed motivation (Theorem 5.1).

Protocol 1: Stimulus Window Scheduling. Rather than continuous access to digital stimuli (which maintains $s \gg s^*$ at all times), stimulus access is confined to defined temporal windows. The rationale follows directly from Proposition 7.2: if s is reduced below s^* for sustained periods, the desensitized receptor parameters ($\mathcal{R}_{\max}$, K_d) partially recover. The recommended structure is:

- Define an *access window* of duration τ_a and a *deprivation window* of duration τ_r (recovery period).
- The recovery period must satisfy $\tau_r > \tau_{\text{recov}}$, where τ_{recov} is the receptor resensitization timescale (empirically, on the order of 24–72 hours for dopaminergic desensitization [Volkow et al., 2011]).
- During deprivation windows, the reduced s allows the Hill-function parameters to shift back toward baseline, restoring sensitivity.

Protocol 2: Latency Injection. The sublimation collapse theorem (Theorem 6.1) establishes that $\kappa \rightarrow 0$ when $\tau_d \rightarrow 0$. The intervention is to deliberately increase the discharge latency τ_d by interposing delays, effort requirements, or procedural friction between impulse and consumption. When τ_d is raised above the sublimation timescale τ_s , the sublimation operator $\mathcal{T}$ regains the opportunity to act on undischarged motivational energy, channeling it into structured activity.

Concrete implementations include: removing instant-access applications from primary devices; imposing mandatory waiting periods before consumption decisions; and restructuring digital environments so that the path to stimuli requires multi-step navigation (increasing the “motivational resistance” R in the Ohmic representation, Equation (12)).

A.2 Stimulus Field Differentiation: Combating Motivational Diffusion

When the stimulus field is approximately uniform ($\mathcal{S}(\mathbf{m}, t) \approx S_0$ for all $\mathbf{m}$), the potential landscape is flat and coherence cannot form (Proposition 4.2). The intervention is to *artificially differentiate* the stimulus field by restricting the number of accessible objects.

Protocol 3: Radical Narrowing. The number of active motivational objects is deliberately reduced to one or two for a defined period. This is not a recommendation born of ascetic philosophy but of the coherence mathematics: with total energy E fixed (Axiom 3.4), concentrating it in fewer regions of $\mathcal{M}$ mechanically increases $\rho_{\max}$ and raises Λ above the critical threshold Λ^* . The sublimation operator (Definition 6.2) then has a well-defined target $\mathbf{m}_0$ around which to concentrate motivational density.

Protocol 4: Contrast Cycling. Alternating between periods of high stimulation and periods of near-complete sensory reduction. The purpose is to recalibrate the baseline of the receptor response function. In the Hill-equation model, if s is periodically reduced to near zero, the system’s operating point on the response curve shifts back to the steep, sensitive region ($s \ll K_d$), restoring the marginal response $d\mathcal{R}/ds$ to its maximum. The subsequent reintroduction of moderate stimuli is then experienced with renewed salience—the same objective stimulus produces a larger subjective response.

A.3 Interpersonal Synchronization Exercises

The Kuramoto model of co-regulation (Section 9) identifies the coupling strength K as the critical parameter determining whether two individuals achieve phase-locked synchronization ($K \geq K_c$) or remain in the drifting regime ($K < K_c$). Interventions should target K directly.

Protocol 5: Rhythmic Entrainment. Joint activities that impose a shared temporal structure—synchronized breathing, rhythmic physical movement, coordinated musical activity—serve to increase the effective coupling strength K between two motivational oscillators. The shared rhythm provides a common external frequency that reduces the effective frequency mismatch $|\omega_1 - \omega_2|$, lowering the synchronization threshold K_c and making phase-locking achievable at weaker coupling strengths.

Protocol 6: Optimal Distance Management. The co-regulation model implies that both extremes—total fusion ($|\omega_1 - \omega_2| = 0$, no dynamic tension) and total separation ($K = 0$, no coupling)—are suboptimal. In the first case, the phase difference $\phi^* = 0$ and the system is synchronized but static, producing no motivational gradient between partners. In the second, no synchronization occurs. The productive regime lies at intermediate coupling: K slightly above K_c , with a residual phase offset $\phi^* = \arcsin(\Delta\omega/2K) > 0$. This offset is experienced as the sustained, productive tension of desire within a stable bond. The practical recommendation is to maintain a balance of shared engagement (high K) and individual autonomy (nonzero $\Delta\omega$).

A.4 Sublimation as Cognitive Architecture: Channeling Undischarged Energy

The sublimation operator (Definition 6.2) transforms high-entropy motivational distributions into low-entropy structured ones. Practical sublimation requires two conditions:

(i) undischarged motivational energy must be available ($\tau_d > \tau_s$), and (ii) a structured activity must be accessible that can serve as the focus $\mathbf{m}_0$.

Protocol 7: The Low-Pass Filter Method. Introduce a daily practice of structured output (writing, drawing, physical training, mathematical work) that acts as a “low-pass filter” on motivational fluctuations: high-frequency, impulsive signals are attenuated by the slow, disciplined structure of the activity, while sustained motivational currents are amplified through coherent channeling. The formal criterion is that the activity’s characteristic timescale τ_{act} must exceed the noise timescale τ_ξ but fall below the frustration threshold at which the system discharges through alternative channels.

Protocol 8: Environmental Redesign. Reconfigure the physical and digital environment to alter the constraint function $\mathcal{C}$ so that destructive discharge paths carry high cost and constructive (sublimatory) paths carry low cost. In the Ohmic representation, this means increasing the “resistance” R on paths to impulsive consumption while decreasing R on paths to structured engagement. Concrete implementations: placing creative tools at hand while placing consumption devices behind barriers; configuring digital environments to open productive applications by default; designing physical spaces that afford focused work and penalize distraction.

B Self-Assessment Tool: Phase Regime Identification

The following instrument is designed to allow an individual to approximate their current position on the phase diagram (Figure 4) by estimating the two primary control parameters—stimulus accessibility (s) and normative constraint density (ν)—and the key derived quantities.

B.1 Stimulus Accessibility Assessment

Rate each item from 1 (strongly disagree) to 5 (strongly agree):

- S1.** I can access entertainment, social media, or stimulating content within seconds at any time of day.
- S2.** I rarely experience a delay between wanting something and obtaining it.
- S3.** I am exposed to a large number of competing stimuli (notifications, content feeds, options) daily.
- S4.** I seldom experience boredom or unstimulated periods lasting more than a few minutes.
- S5.** New sources of stimulation are always readily available when current ones lose their appeal.

Scoring: Sum the responses. A total of 20–25 indicates $s \gg s^*$ (deep saturation regime). A total of 10–19 indicates moderate accessibility. A total of 5–9 indicates low accessibility (possible repressive regime).

B.2 Normative Constraint Assessment

Rate each item from 1 (strongly disagree) to 5 (strongly agree):

- C1. I experience strong social, moral, or institutional prohibitions on many of my desires.
- C2. I regularly suppress impulses because of fear of judgment or punishment.
- C3. My environment imposes rigid expectations about what I should want or pursue.
- C4. I feel that most paths of action are blocked or heavily penalized.
- C5. I experience significant guilt or anxiety when I deviate from prescribed behavior.

Scoring: A total of 20–25 indicates $\nu \gg 1$ (deep repressive regime). A total of 10–19 indicates moderate constraint. A total of 5–9 indicates low constraint.

B.3 Coherence Indicator

Rate each item from 1 (strongly disagree) to 5 (strongly agree):

- $\Lambda 1$. I have a clear, sustained focus on one or two deeply engaging projects, relationships, or goals.
- $\Lambda 2$. I can maintain concentrated interest in a single activity for extended periods without seeking distraction.
- $\Lambda 3$. My desires feel directed and purposeful rather than scattered or diffuse.
- $\Lambda 4$. I experience deep satisfaction (not merely momentary pleasure) from my primary engagements.
- $\Lambda 5$. I choose not to pursue many available options because my current focus is sufficiently absorbing.

Scoring: A total of 20–25 indicates $\Lambda \gg \Lambda^*$ (high coherence—Regime II). A total below 10 indicates motivational diffusion (Regime III or IV).

B.4 Interpretation Guide

Table 4: Self-assessment interpretation matrix.

s score	ν score	Likely regime	Primary intervention direction
Low	High	I (Repression)	Reduce constraint density; increase access to moderate, chosen stimuli
Moderate	Moderate	II (Coherent)	Maintain current balance; optimize sublimation channels
High	Low	III (Saturation)	Reduce stimulus access; inject latency; narrow focus
High	Low, with low Λ over prolonged period	IV (Anhedonia)	Professional support; graduated stimulus reintroduction; co-regulation

This instrument is not a clinical diagnostic tool but a heuristic device grounded in the framework’s formal structure. Its purpose is to translate the abstract parameters of the phase diagram into assessable experiential indicators, enabling individuals to identify which control parameters require modification and in which direction.